This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



# SSCD-HI: Intervention-Guided Spectral–Causal Representation Learning for Hyperspectral Image Classification

Xiaofei Yang *Member, IEEE,* Tong Tang, Ying Liu *Member, IEEE*, Yuanchao Su *Senior Member, IEEE*, Yifang Ban *Senior Member, IEEE*, and Yicong Zhou *Senior Member, IEEE*

***Abstract*—Hyperspectral image classification under sparse supervision is susceptible to correlations between land-cover labels and nuisance factors such as illumination, background context, and spectral mixing. Learning expressive spectral–spatial features does not by itself determine which of these correlations a classifier should retain. This article proposes spectral–causal disentanglement with hierarchical intervention (SSCD-HI), a framework that constrains latent representations through their roles in prediction. Adaptive masks organize encoded features into two components intended to carry label-stable and nuisance-sensitive information. Hierarchical intervention exchanges nuisance-sensitive features between real samples while retaining the recipient's label-stable representation and supervision. Within-class exchange encourages consistency under label-compatible variation, whereas cross-class exchange challenges reliance on nuisance cues associated with competing classes. A complementary decorrelation objective limits redundancy between the latent components. These operations reuse encoded features during training and introduce no additional intervention path at inference. Experiments on Indian Pines, Houston2013, and WHU-Hi-LongKou show improved overall and average classification accuracy over the evaluated baselines under limited-label settings. Component ablations assess the training objectives, while perturbation analyses reveal scene-dependent sensitivity to spectral noise. Branch diagnostics distinguish reduced predictive reliance from removal of class information. The results support intervention-guided regulation of latent information as a complementary approach to increasing backbone capacity for label-efficient hyperspectral classification.**

***Index Terms*—Hyperspectral image classification, causal representation learning, latent intervention, nuisance invariance, spectral–spatial representation.**

(Corresponding authors: Ying Liu and Yuanchao Su.)

This work was jointly supported by the following projects: National Natural Science Foundation of China (NSFC) Fund under Grant 62301174. This work was supported in part by the National Natural Science Foundation of China under Grant 42571463, in part by the Natural Science Foundation Research Program of Shaanxi Province under Grant 2025JC-YBMS-257, in part by the Open Research Fund of Shaanxi Key Laboratory of Optical Remote Sensing and Intelligent Information Processing under Grant KF20250302.

Xiaofei Yang, Tong Tang and Ying Liu are with the School of Electronic and Communication Engineering, University Name, Guangzhou University, Guangzhou 510182, China. (e-mail: xiaofeiyang@gzhu.edu.cn, 32407600002@e.g2hu.edu.cn, amy.yingliu@gzhu.edu.cn).

Yuanchao Su is with the College of Geomatics, Xi'an University of Science and Technology, Xi'an 710054, China, and also with Shaanxi Yellow River Planning and Design Company Ltd., Yulin 719000, China (e-mail: suych3@xust.edu.cn).

Yifang Ban is with the Division of Geoinformatics, School of Architecture and the Built Environment, KTH Royal Institute of Technology, 11428 Stockholm, Sweden (e-mail: yifang@kth.se).

Yicong Zhou is with the Department of Computer and Information Science, University of Macau, Macau, China (e-mail: yicongzhou@um.edu.mo).

## I. Introduction

HYPERSPECTRAL imaging (HSI) provides dense spectral measurements that enable fine discrimination of materials with similar visual appearance, supporting applications such as land-cover analysis, precision agriculture, mineral exploration, and environmental monitoring [1], [2]. Despite this rich spectral information, reliable HSI classification remains difficult when labeled samples are limited. The observed spectrum of a material is shaped not only by its intrinsic properties but also by illumination, local background, spectral mixing, and sensor perturbations. These nuisance factors can become correlated with class labels in the training set, allowing a classifier to exploit predictive but unstable evidence. Such dependence is particularly problematic for spectrally similar or sparsely represented categories, where nuisance-sensitive cues can easily dominate the limited class-specific information.

Modern architectures have substantially improved spectral–spatial representation learning. CNNs capture local spatial structure and spectral continuity [3]. Transformer models extend contextual reasoning through self-attention [4]–[6], and recent state-space models provide efficient long-range spectral–spatial dependency modeling [7]–[9]. These advances improve what a classifier can represent, but not necessarily what information it relies on. A larger receptive field may propagate material-related evidence and nuisance correlations simultaneously. The central difficulty is therefore not solely how to encode richer spectral–spatial context, but how to distinguish label-stable information from nuisance-sensitive variation and regulate their respective effects on prediction.

Causality-inspired HSI methods offer a relevant perspective. Existing studies have explored causal meta-learning for domain generalization, disentangled representations for cross-scene recognition, counterfactual augmentation, and causal constraints in attention-based models [10]–[13]. Their emphasis, however, is largely placed on cross-scene invariance, synthesized counterfactual variation, or architecture-specific causal reasoning. In standard single-scene classification, nuisance information may already be embedded in the learned spectral–spatial representation. Merely splitting this representation into multiple branches does not ensure functional disentanglement, because different branches may retain overlapping class or nuisance information. A meaningful decomposition therefore requires the latent factors to be actively perturbed so that their roles in the classification decision are constrained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



during learning.

Motivated by this observation, we propose spectral–causal disentanglement with hierarchical intervention (*SSCD-HI*). Adaptive spectral–causal masks first factorize patch-level features into label-stable and nuisance-sensitive components without imposing a fixed channel partition. Here, label-stable and nuisance-sensitive name intended functional roles. The historical terms causal and spurious are retained in the method name and superscripts. Hierarchical latent intervention then regulates the use of these components: within-class nuisance exchange encourages prediction stability under admissible intra-class variation, whereas cross-class intervention discourages excessive dependence on nuisance-sensitive information. A latent alignment penalty discourages coordinate-wise alignment between the two components. Importantly, these intervention paths reuse encoded features only during training and are removed at inference, leaving the standard encoder–classifier prediction path unchanged.

The main contributions of this work are summarized as follows:

- We formulate SSCD-HI as an intervention-guided spectral–causal representation framework that adaptively separates label-stable and nuisance-sensitive latent factors and regulates their predictive roles through hierarchical real-sample intervention. The defining operation is label-preserving substitution within a designated latent component, with separate within-class and cross-class donor rules.
- Adaptive spectral–causal factorization is realized through learnable masks that organize each patch-level representation into label-stable and nuisance-sensitive components without imposing a fixed channel partition.
- We introduce hierarchical latent intervention with within-class nuisance exchange and cross-class feature recombination, together with an angular alignment penalty, to regulate the functional roles of the two latent components.
- Experiments on three HSI benchmarks assess classification performance, component contributions, spatial context, label availability, and spectral perturbations. Additional branch diagnostics examine how information is represented and used.

## II. Related Work

### A. Spectral–Spatial Representation Learning

HSI classification has progressed through increasingly expressive spectral–spatial encoders. CNN-based models exploit locality and spectral continuity, with hybrid 3-D/2-D designs remaining effective for compact representation learning [3], [14]. Transformers extend the receptive field through attention: SpectralFormer emphasizes inter-band dependency [4], SSFTT introduces spectral–spatial tokenization [15], group-aware hierarchical attention organizes local token groups [16], and dual-branch or factorized designs balance local structure with global context [6], [17], [18]. State-space models offer another route to long-range modeling by adapting selective scanning to spectral, spatial, or joint structures [7], [9], [19]–[21].

These advances enlarge what can be represented, but they do not determine which predictive cues should be retained. Under empirical risk minimization, material-related evidence and contextual correlations are equally useful if both reduce training error. A CNN may exploit a recurring local background, while attention or state-space propagation can carry the same cue over a broader context. The limitation considered here is therefore not receptive-field size or backbone capacity, but the absence of a mechanism that distinguishes label-stable information from nuisance-sensitive evidence once both have entered the latent representation.

### B. Nuisance Variability and Shortcut Learning in Hyperspectral Data

This ambiguity is inherent to hyperspectral observation. Spectral responses vary with atmospheric state, illumination and viewing geometry, topography, and local environmental conditions even within a single acquisition [22]. Sub-pixel mixing further couples a pixel to scene-dependent endmember proportions [23]. Because these factors are spatially structured rather than independent of the label, a single-scene training set can contain systematic class–context associations. Empirical optimization can then favor an accessible nuisance cue over the more stable material evidence, consistent with shortcut learning [24].

The resulting error is often class dependent. Dominant categories may remain recognizable from their aggregate spectral profile, whereas spectrally similar or sparsely labeled classes are more sensitive to displaced class-specific evidence. High overall accuracy can therefore coexist with weaker class-balanced performance. More importantly, this failure cannot be identified from training risk alone. It motivates objectives that constrain invariance or causal stability rather than only increasing model capacity [25]–[27].

### C. Causality-Inspired Representation Learning and Its Scope

Causality-inspired HSI methods mainly address invariance across domains or impose causal structure through a particular architecture. Related domain-learning approaches also regulate spectral–spatial evidence: Huang et al. use two-branch attention with adversarial domain adaptation for cross-domain HSI classification [28], whereas DADAnet constructs adversarially augmented domains for single-domain generalization [29]. These methods address transfer across scenes. SSCD-HI studies within-scene latent substitution, retaining the label of the sample that supplies the label-stable component. Causal meta-generalization learns transferable knowledge from virtual domains [10]. Explicit semantic guidance and causal inference reduce scene-specific dependence in cross-scene classification [30]–[32], and counterfactual augmentation supports hyperspectral domain generalization [12]. Such formulations define stability across scenes and therefore rely on multiple domains or mechanisms that synthesize domain variation [25], [33].

A second line separates transferable and domain-related information structurally [11], [34], [35], or embeds causal reasoning into a specific mechanism, as in causal attention [13], [36]. Structural separation alone, however, does not

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



assign a unique function to each branch: without an additional inductive signal, multiple latent factors may encode overlapping evidence [37]. Architecture-specific causal modules, in turn, tie the causal argument to the mechanism in which it is implemented.

SSCD-HI targets a different setting: training and inference occur within one scene, while nuisance-sensitive information is already entangled with class evidence in the latent representation. Adaptive masks organize that representation into label-stable and nuisance-sensitive components, and hierarchical intervention replaces the nuisance-sensitive component with that of another real sample while retaining the original supervision. The latent split is therefore constrained by its effect on prediction rather than by branch structure alone. This operation also differs from Mixup and manifold Mixup, which interpolate both samples and labels [38], [39]. SSCD-HI performs a hard substitution in a designated latent subspace and preserves the original label. Finally, although kernel- and distance-based criteria can suppress nonlinear dependence [40], [41], the proposed framework uses lightweight normalized alignment only to reduce residual redundancy, because the intervention itself provides the functional constraint on the two latent roles.

## III. Proposed Method

### A. Problem Formulation and Causal Assumptions

Let an HSI cube be denoted by $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where $H$, $W$, and $C$ represent the spatial height, spatial width, and number of spectral bands, respectively. Given a label map $\mathbf{Y} \in \{0, 1, \ldots, K\}^{H\times W}$, with 0 denoting unlabeled background pixels, a labeled pixel at $(u_i, v_i)$ is represented by a local spectral–spatial patch

$$\mathbf{P}_i = \mathcal{N}_p(\mathbf{X}, u_i, v_i) \in \mathbb{R}^{C\times p\times p}, \tag{1}$$

where $p$ is the patch size and $y_i \in \{1, \ldots, K\}$ is its class label. The conventional objective is to learn $f_\Theta : \mathbf{P}_i \mapsto y_i$ by minimizing empirical classification risk.

*1) Generative View:* Empirical risk minimization is agnostic to *why* a response is predictive, so the difficulty described in Section II cannot be expressed in terms of $f_\Theta$ alone. We therefore make the observation process explicit. Let $Y$ denote the material class of the center pixel and let $U$ collect the observation-dependent state of the acquisition: illumination and viewing geometry, atmospheric condition, local topography, neighborhood composition, sub-pixel mixing proportions, and sensor perturbation. A patch is generated as

$$\mathbf{P} = h\left(\mathbf{s}(Y),\, U,\, \boldsymbol{\varepsilon}\right), \tag{2}$$

where $\mathbf{s}(\cdot)$ maps a class to its intrinsic spectral signature, $h(\cdot)$ is the unknown radiometric and spatial observation operator, and $\boldsymbol{\varepsilon}$ is label-independent noise. Equation (2) restates a standard property of hyperspectral data: the recorded signature of a fixed material is modulated by conditions that vary within one acquisition [22], [23].

The decisive point is the relation between $Y$ and $U$. Over a physically plausible population of scenes these are distinct causes of $\mathbf{P}$. Within one annotated scene, however, spatial layout makes them dependent: a crop variety may occupy one slope aspect, and one urban material may border one recurring background. The training distribution therefore satisfies

$$p_{\mathrm{tr}}(U \mid Y) \neq p_{\mathrm{tr}}(U), \tag{3}$$

as depicted in Fig. 2(a). Under (3), some statistics of $U$ may predict $Y$ on the training set, so minimizing classification risk alone need not reject nuisance-dependent cues. This is the formal content of the shortcut mechanism discussed in Section II.

*2) Target Property:* We do not attempt to recover $\mathbf{s}(\cdot)$, $h(\cdot)$, or $U$. The objective is weaker and directly testable: the prediction should be unchanged when the observation state is replaced while the material is held fixed,

$$f_\Theta\left(h(\mathbf{s}(y), u, \boldsymbol{\varepsilon})\right) = f_\Theta\left(h(\mathbf{s}(y), u', \boldsymbol{\varepsilon})\right) \quad \forall\, u, u' \in \mathcal{U}_y, \tag{4}$$

where $\mathcal{U}_y$ is the set of observation states under which class $y$ can plausibly be acquired. Condition (4) describes an ideal physical invariance under $\mathrm{do}(U := u')$, illustrated in Fig. 2(b). Its image-domain evaluation would require physically valid observations of the same material under different acquisition states. SSCD-HI instead substitutes learned latent components. This operation is a representation-level training perturbation. Correspondence to a physical intervention requires additional assumptions.

Accordingly, we express the latent descriptor as

$$\mathbf{z}_i = \psi\left(\mathbf{z}_i^c, \mathbf{z}_i^s\right), \tag{5}$$

and adopt three assumptions.

- **A1** *(Latent separability)* There exists a factorization of $\mathbf{z}_i$ under which $\mathbf{z}_i^c$ depends on $\mathbf{P}_i$ mainly through $\mathbf{s}(Y)$ and $\mathbf{z}_i^s$ mainly through $U$.
- **A2** *(Nuisance transferability)* If $\mathbf{z}_j^s$ encodes state $u_j$, then $u_j \in \mathcal{U}_{y_i}$ for the classes $y_i$ present in the same scene, so a recombined descriptor corresponds to an observation that could have occurred.
- **A3** *(Slot-wise action)* The classifier consumes the two components through distinct input coordinates, so replacing one leaves the other unaltered.

A1 and A2 are modeling assumptions rather than properties guaranteed by the objective. A3 specifies the concatenated input coordinates and can be checked from the implementation, although a nonlinear head can couple those coordinates. The superscripts $c$ and $s$ denote intended roles rather than identified physical factors. A2 is the assumption most exposed to violation, since an observation state typical of a dense urban block is a poor surrogate for a crop pixel.

### B. Overview of SSCD-HI

Fig. 1 shows the framework. A compact spectral–spatial encoder maps each patch to a latent descriptor, and adaptive spectral–causal masks organize that descriptor into the two components of (5). Because A1 is an unverified modeling assumption, the masks alone establish nothing: two separately parameterized branches may still encode the same evidence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



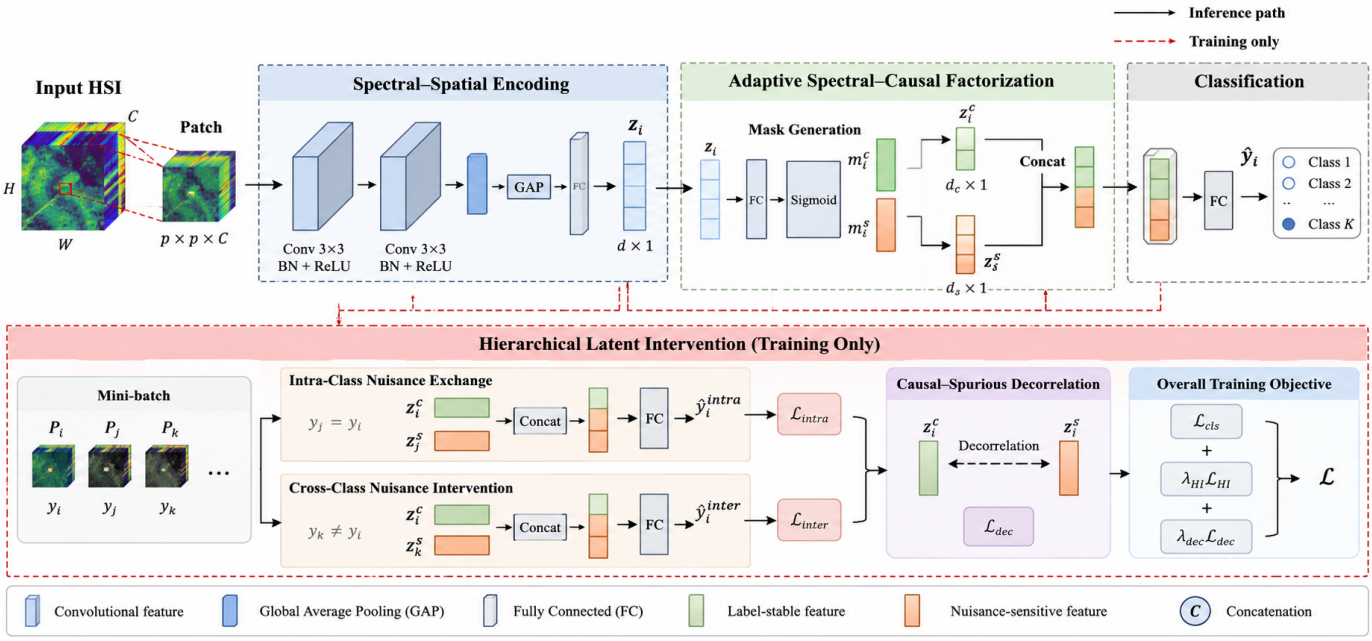
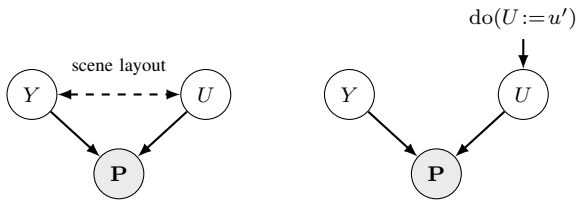



Fig. 1: Overview of the proposed SSCD-HI framework. An input HSI patch is first encoded into a compact spectral–spatial representation and adaptively factorized into label-stable and nuisance-sensitive components. The solid black path denotes the standard inference procedure. During training, intra-class nuisance exchange and cross-class nuisance intervention regulate the functional roles of the two latent components, while an angular alignment penalty discourages coordinate-aligned branch features. The intervention branches reuse encoded latent features and are removed during inference.




(a) training scene (b) after intervention

Fig. 2: Assumed data-generating process for a labeled HSI patch. (a) In a single annotated scene the material class $Y$ and the observation state $U$ are distinct causes of the patch $\mathbf{P}$, but spatial layout renders them statistically dependent (dashed link), so some statistics of $U$ may predict $Y$. (b) Assigning $U$ externally removes that dependence. Panel (b) depicts the physical target. SSCD-HI uses a latent substitution motivated by this target. Its correspondence to an acquired counterfactual patch is conditional on A1 and A2.

SSCD-HI therefore adds constraints that act on the *use* of the components. Hierarchical latent intervention substitutes $\mathbf{z}^s$ during training, and the alignment penalty discourages parallel or antiparallel branch features.

Two paths follow. The classification path, drawn as the solid route in Fig. 1, consumes the complete factorized descriptor and is active in both training and inference. The intervention path, drawn as the branching routes, is active only during training and is built from features the encoder has already produced. The training signal is thus modified without a second encoder pass, and inference is unaffected.

### C. Spectral–Spatial Encoding

The framework is intended to isolate the effect of factorization and intervention, so the backbone is deliberately kept minimal: any gain that a heavier encoder might contribute would confound the attribution. For an input patch $\mathbf{P}_i$, two consecutive $3 \times 3$ convolutional blocks produce

$$\mathbf{F}_i^{(1)} = \sigma\left[\mathrm{BN}_1\left(\mathrm{Conv}_{3\times3}(\mathbf{P}_i)\right)\right], \tag{6}$$

and

$$\mathbf{F}_i^{(2)} = \sigma\left[\mathrm{BN}_2\left(\mathrm{Conv}_{3\times3}(\mathbf{F}_i^{(1)})\right)\right], \tag{7}$$

where $\sigma(\cdot)$ denotes ReLU activation and $\mathrm{BN}(\cdot)$ denotes batch normalization. Spatial responses are aggregated by global average pooling (GAP) and linearly projected,

$$\mathbf{z}_i = \sigma\left[\mathbf{W}_e\,\mathrm{GAP}\left(\mathbf{F}_i^{(2)}\right) + \mathbf{b}_e\right], \qquad \mathbf{z}_i \in \mathbb{R}^d. \tag{8}$$

All subsequent operations act on $\mathbf{z}_i$ only. The encoder is consequently an interface rather than a component of the contribution, and the latent operators can be attached to other encoders with compatible output dimensions. Their effectiveness with Transformer or state-space encoders is not evaluated here.

### D. Adaptive Spectral–Causal Factorization

Sample-dependent masks modulate two learned projections of the same descriptor. This allows the relative contribution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



of each projected coordinate to vary across patches. The mask generator is

$$[\mathbf{m}_i^c, \mathbf{m}_i^s] = \mathrm{Split}\left\{\sigma_g\left(\mathbf{W}_m \mathbf{z}_i + \mathbf{b}_m\right)\right\}, \tag{9}$$

where $\mathbf{W}_m \in \mathbb{R}^{(d_c+d_s)\times d}$ and $\mathbf{b}_m \in \mathbb{R}^{d_c+d_s}$. The operation Split slices the first $d_c$ outputs into $\mathbf{m}_i^c$ and the remaining $d_s$ outputs into $\mathbf{m}_i^s$. Each element is independently transformed by the sigmoid $\sigma_g$. In the compact model, $d$=128 and $d_c = d_s$=64. Thus, $\mathbf{m}_i^c, \mathbf{m}_i^s \in (0,1)^{64}$. No constraint enforces $\mathbf{m}_i^c + \mathbf{m}_i^s = \mathbf{1}$.

The components are

$$\mathbf{z}_i^c = (\mathbf{W}_c \mathbf{z}_i + \mathbf{b}_c) \odot \mathbf{m}_i^c, \tag{10}$$

and

$$\mathbf{z}_i^s = (\mathbf{W}_s \mathbf{z}_i + \mathbf{b}_s) \odot \mathbf{m}_i^s, \tag{11}$$

where $\odot$ denotes element-wise multiplication. The two masks share an input descriptor but have separate rows in $\mathbf{W}_m$. Shared generation does not impose competition, exclusivity, or a complementary partition. Overlap is possible, and the alignment penalty does not guarantee removal of label information from either branch. Both branches use the same width, $d_c = d_s$, which permits the direct alignment penalty of Section III-E2.

The classification path retains both components,

$$\mathbf{o}_i = g_\phi\left([\mathbf{z}_i^c, \mathbf{z}_i^s]\right), \tag{12}$$

where $[\cdot,\cdot]$ denotes concatenation. The compact implementation uses a nonlinear head: a 128-to-256 linear layer, batch normalization, ReLU, and a 256-to-$K$ output layer. Both components are retained at inference because label information may remain in either branch. Learned projections and sigmoid gates are not assumed invertible, so concatenation does not guarantee an information-preserving factorization.

### E. Hierarchical Latent Intervention

Given a mini-batch

$$\mathcal{B} = \{(\mathbf{P}_i, y_i)\}_{i=1}^B, \tag{13}$$

we construct latent pairs whose target is the label attached to the retained label-stable component. The two donor rules below sample within and across classes. They regularize prediction under latent recombination. A physically realizable recipient observation follows only if A1 and A2 hold. Donors are drawn uniformly over eligible samples in the mini-batch, without a spectral-distance or material-compatibility filter. Consequently, a class with more eligible samples contributes more donors. A donor drawn from the same scene is not automatically physically compatible with every recipient class.

*1) Intra-Class Nuisance Exchange:* Samples of one land-cover category differ in illumination, local background, mixing proportion, canopy density, and neighborhood heterogeneity. Drawing the donor from the same class samples $u'$ from $p_{\mathrm{tr}}(U \mid Y = y_i)$, which samples observed within-class variability rather than the entire physically admissible set $\mathcal{U}_{y_i}$. When another same-class sample is present, an index $\pi_{\mathrm{same}}(i)$ is selected such that

$$y_{\pi_{\mathrm{same}}(i)} = y_i, \qquad \pi_{\mathrm{same}}(i) \neq i. \tag{14}$$

The label-stable component is retained and the nuisance-sensitive component is replaced,

$$\widetilde{\mathbf{z}}_i^{\mathrm{intra}} = \left[\mathbf{z}_i^c, \mathbf{z}_{\pi_{\mathrm{same}}(i)}^s\right], \tag{15}$$

and supervised by the original label,

$$\mathcal{L}_{\mathrm{intra}} = \frac{1}{B}\sum_{i=1}^B \mathrm{CE}\left(g_\phi(\widetilde{\mathbf{z}}_i^{\mathrm{intra}}), y_i\right). \tag{16}$$

The target is unchanged because the recipient supplies $\mathbf{z}_i^c$. Same-class pairing reduces direct label conflict but does not certify physical compatibility or eliminate class information in $\mathbf{z}^s$. If a class appears once, the implementation uses the sample itself as donor: this term then reduces to ordinary classification for that sample.

*2) Cross-Class Nuisance Intervention:* Stability alone is insufficient. A classifier that reads $y_i$ off $\mathbf{z}_i^s$ would satisfy (16) perfectly, because within-class donors carry the same label. Breaking the association in (3) requires donors drawn in conflict with the label. When the batch contains at least two classes, for each $i$ an index $\pi_{\mathrm{cross}}(i)$ satisfies

$$y_{\pi_{\mathrm{cross}}(i)} \neq y_i, \tag{17}$$

giving

$$\widetilde{\mathbf{z}}_i^{\mathrm{inter}} = \left[\mathbf{z}_{\pi_{\mathrm{cross}}(i)}^c, \mathbf{z}_i^s\right], \tag{18}$$

with the target $y_{\pi_{\mathrm{cross}}(i)}$ attached to the stable-component supplier,

$$\mathcal{L}_{\mathrm{inter}} = \frac{1}{B}\sum_{i=1}^B \mathrm{CE}\left(g_\phi(\widetilde{\mathbf{z}}_i^{\mathrm{inter}}), y_{\pi_{\mathrm{cross}}(i)}\right). \tag{19}$$

The target follows the component designated label-stable. The implementation equivalently constructs $[\mathbf{z}_j^c, \mathbf{z}_i^s]$ with target $y_j$, where $j$ is drawn from samples whose label differs from $y_i$. Here $j$ is the recipient of the nuisance component, even though it is sampled as the stable-component donor in code. This distinction specifies the direction of supervision and avoids interpreting the nuisance supplier's label as the target. Because cross-class eligibility depends on labels, the donor branch need not be statistically independent of the target.

**Remark 1.** *For a linear head,* $\mathbf{o} = \mathbf{W}_c\mathbf{z}^c + \mathbf{W}_s\mathbf{z}^s + \mathbf{b}$*, the contribution of a recombined example to the gradient with respect to* $\mathbf{W}_s$ *is*

$$(\mathrm{softmax}(\mathbf{o}) - \mathbf{e}_{y_i})(\mathbf{z}_j^s)^\top.$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



*This expression does not imply a class-independent nuisance contribution at stationarity. Conditional on a donor from class $a$, even class-uniform cross-class pairing gives target probabilities zero for $a$ and $1/(K-1)$ for every other class. For $K$=2, the donor label determines the opposite target exactly, so an informative nuisance branch can support cross-class prediction. The loss therefore encourages recipient-label consistency over sampled recombinations without guaranteeing nuisance neutrality. This limitation also applies to the nonlinear head used here.*

Within-class exchange and cross-class intervention impose different training conditions: the former samples label-compatible donors, whereas the latter tests prediction when the nuisance supplier has a competing label. Neither condition establishes causal identification or statistical independence of the two components.

They are combined as

$$\mathcal{L}_{\mathrm{HI}} = \frac{1}{2}\left(\mathcal{L}_{\mathrm{intra}} + \mathcal{L}_{\mathrm{inter}}\right). \tag{20}$$

The cost of (19) should be stated plainly. It is the term that depends on A2, and A2 weakens as the donor becomes less plausible for the recipient class. Where the assumption fails, the recombined descriptor can lie outside the support of recipient-class observations, and the associated gradient can bias learning. This is why $\mathcal{L}_{\mathrm{HI}}$ enters the objective with a finite weight instead of as a constraint, and motivates examining sensitivity to its weight.

An angular penalty complements the intervention objective.

The intervention loss acts on predictions from recombined features. An additional alignment penalty discourages parallel or antiparallel representations but does not enforce statistical independence.

With $d_c = d_s$, SSCD-HI penalizes their normalized alignment,

$$\mathcal{L}_{\mathrm{dec}} = \frac{1}{B}\sum_{i=1}^{B}\left(\frac{(\mathbf{z}_i^c)^\top \mathbf{z}_i^s}{\|\mathbf{z}_i^c\|_2\|\mathbf{z}_i^s\|_2 + \epsilon}\right)^2, \tag{21}$$

where $\epsilon > 0$ provides numerical stability and the square makes the penalty sign-invariant, so that anti-aligned duplication is discouraged as well. Equation (21) measures within-sample angular alignment. Orthogonality in these coordinates can coexist with shared label information or nonlinear dependence, so the penalty should not be interpreted as an independence test. Kernel- or distance-based dependence measures [40], [41] assess different properties and can require quadratic batch-wise computation.

### F. Degenerate Solutions and the Role of Each Constraint

The objective admits solutions that differ from the intended two-role interpretation. If $\mathbf{z}^c$ is uninformative, cross-class training creates conflicting demands on the remaining branch. However, the label-conditioned pairing in Remark 1 can itself retain predictive information. Thus, the loss does not generally exclude a weak label-stable branch. If the branches duplicate class information, the alignment penalty discourages coordinate-aligned duplication but can miss information encoded after a rotation or nonlinear transformation. Cross-class substitution of a duplicate can also introduce conflicting class evidence, so its effect is not vacuous.

If $\mathbf{z}^s$ collapses to zero, the two intervention losses reduce to classification on $\mathbf{z}^c$ and the alignment penalty vanishes. The objective permits this solution. Component accuracy ablations alone cannot distinguish it from a noncollapsed, actively regulated branch. Branch norms, held-out probes, and fixed-head substitutions assess these different possibilities. The response letter reports the available diagnostics.

### G. Learning Objective

Classification on the unmodified representation is optimized using

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{B}\sum_{i=1}^{B}\mathrm{CE}\left(g_\phi([\mathbf{z}_i^c, \mathbf{z}_i^s]), y_i\right), \tag{22}$$

and the complete objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{HI}}\mathcal{L}_{\mathrm{HI}} + \lambda_{\mathrm{dec}}\mathcal{L}_{\mathrm{dec}}. \tag{23}$$

The three terms are not interchangeable and their weights are not interchangeable either. $\mathcal{L}_{\mathrm{cls}}$ anchors the representation to the observed data and prevents the intervention terms from being satisfied by an uninformative encoder. $\lambda_{\mathrm{HI}}$ sets how far the objective is willing to move toward (4) at the risk of the off-support gradients discussed in Section III-E, so it is the parameter that trades invariance against fidelity to the observed distribution. $\lambda_{\mathrm{dec}}$ controls angular alignment, rather than removal of all shared information. A large value would force an artificial orthogonality that the generative model in (2) does not warrant, since $\mathbf{s}(Y)$ and $U$ interact in $h(\cdot)$ and their latent images need not be orthogonal. No auxiliary reconstruction or generative objective is introduced.

### H. Training and Inference

Algorithm 1 summarizes the procedure. Each mini-batch is encoded once. The resulting components are reused for ordinary classification, for both recombinations, and for decorrelation. The pairings $\pi_{\mathrm{same}}$ and $\pi_{\mathrm{cross}}$ are drawn from the labels already present in the batch, so the interventions require no sampling from the full training set and no memory bank. A class appearing once uses self-pairing for the intra-class loss. If the batch contains only one class, the cross-class loss also falls back to self-pairing. Both losses remain averaged over all $B$ samples. The substitution can be inactive even though its classification term is evaluated.

At inference the intervention operators are removed, as indicated by the solid route in Fig. 1. A patch is encoded and factorized by (8)–(11) and classified by (12). No pairing, batch statistics beyond normalization, or donor sample is needed, so the deployed model is an ordinary encoder–classifier and predictions are independent across pixels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



**Algorithm 1** Training Procedure of SSCD-HI

**Require:** HSI cube $\mathbf{X}$, label map $\mathbf{Y}$, patch size $p$, batch size $B$, coefficients $\lambda_{\mathrm{HI}}$ and $\lambda_{\mathrm{dec}}$
**Ensure:** Trained model parameters $\Theta$
Sample $\mathcal{B} = \{(\mathbf{P}_i, y_i)\}_{i=1}^{B}$ from labeled pixels
Encode each patch to obtain $\mathbf{z}_i$ by (6)–(8)
Generate $\mathbf{m}_i^c$, $\mathbf{m}_i^s$ by (9)
Compute $\mathbf{z}_i^c$, $\mathbf{z}_i^s$ by (10)–(11)
Compute $\mathcal{L}_{\mathrm{cls}}$ by (22)
Construct same-class mapping $\pi_{\mathrm{same}}(i)$
Form $\widetilde{\mathbf{z}}_i^{\mathrm{intra}}$ and compute $\mathcal{L}_{\mathrm{intra}}$ by (16)
Construct cross-class mapping $\pi_{\mathrm{cross}}(i)$
Form $\widetilde{\mathbf{z}}_i^{\mathrm{inter}}$ and compute $\mathcal{L}_{\mathrm{inter}}$ by (19)
Compute $\mathcal{L}_{\mathrm{HI}}$ by (20) and $\mathcal{L}_{\mathrm{dec}}$ by (21)
Update $\Theta$ by minimizing (23)

### I. Computational Complexity

Let $d_1$ and $d_2$ denote the output widths of the two convolutional blocks. Ignoring constant kernel factors, the encoder dominates the per-sample cost,

$$\mathcal{O}\left(p^2 C d_1 + p^2 d_1 d_2\right), \tag{24}$$

while mask generation and the factorized projections act in the latent space at

$$\mathcal{O}\left(d(d_c + d_s)\right). \tag{25}$$

The interventions add no encoder evaluation. Their cost is two additional passes through the classifier head, $\mathcal{O}(B((d_c + d_s)h + hK))$ per batch for a head of hidden width $h$, plus $\mathcal{O}(B^2)$ for the per-sample scans of batch labels in the current implementation and $\mathcal{O}(B(d_c+d_s))$ for (21). The interventions reuse encoder outputs, but their measured training overhead depends on the head, batch size, and pairing implementation.

At inference the intervention terms are absent altogether, and the only operations beyond the compact encoder are (9), (10), (11), and the head. The proposed mechanism therefore alters the training objective without changing the deployed computation.

## IV. Experiments and Results

### A. Experimental Setup

*1) Datasets and Evaluation Protocol:* Experiments are conducted on three widely used hyperspectral benchmarks with different scene characteristics. Indian Pines is an agricultural scene containing multiple crop and vegetation categories, with substantial spectral similarity between several classes and heterogeneous mixed regions. Following the adopted protocol, 10% of the labeled samples from each class are used for training and the remaining samples are used for testing [42]. Houston2013 represents a more complex urban environment containing vegetation, soil, water, residential areas, roads, railways, parking lots, and other man-made surfaces. The training set contains 1% of the labeled samples [43]. WHU-Hi-LongKou contains agricultural crops together with artificial surfaces, including roads, roofs, and bridges. We use 0.1% of its labeled samples for training [44].

All hyperspectral cubes are normalized to $[0, 1]$, and background pixels are excluded from both training and evaluation. Unless otherwise specified, the input patch size is set to $15 \times 15$. Each main experiment is repeated over ten random runs, and the reported results are expressed as mean $\pm$ standard deviation. The three datasets therefore provide complementary evaluation conditions ranging from agricultural scenes with spectrally similar crops to heterogeneous urban environments.

*2) Evaluation Metrics:* Classification performance is measured using overall accuracy (OA), average accuracy (AA), and Cohen's kappa coefficient. OA reflects the proportion of correctly classified test samples over the complete test set, whereas AA gives equal weight to individual categories and is less dominated by classes with large sample populations. The kappa coefficient measures prediction agreement after accounting for agreement expected by chance. Given a confusion matrix whose entry $n_{ij}$ denotes the number of samples from class $i$ assigned to class $j$, the three metrics are computed as

$$\mathrm{OA} = \frac{\sum_i n_{ii}}{\sum_i \sum_j n_{ij}}, \tag{26}$$

$$\mathrm{AA} = \frac{1}{K} \sum_{i=1}^{K} \frac{n_{ii}}{\sum_j n_{ij}}, \tag{27}$$

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{28}$$

where $p_o$ and $p_e$ denote the observed and chance-expected agreement, respectively. Reporting all three measures is important here because some datasets contain categories that differ markedly in classification difficulty, which may not be fully reflected by OA alone.

*3) Implementation Details and Compared Methods:* SSCD-HI is optimized using AdamW with an initial learning rate of $10^{-3}$ and a weight decay of $10^{-4}$. Cosine annealing is adopted for learning-rate scheduling. The manuscript protocol specifies 100 epochs for Indian Pines and 200 epochs for Houston2013. The archived WHU-Hi-LongKou results matching the main table were trained for 200 epochs. The mini-batch size is 100. The dimensions of the label-stable and nuisance-sensitive representations are both set to 64. The main-protocol loss coefficients are $\lambda_{\mathrm{HI}}$=0.5 and $\lambda_{\mathrm{dec}}$=0.1. Patch size, branch widths, optimizer, and these coefficients are shared. The epoch budget is dataset dependent. Supplemental runs below use explicitly separate configurations. Random flipping, mild radiation perturbation, and mixture augmentation are applied during training.

The comparison covers three representative families of HSI classifiers. The CNN group contains 2D-CNN, 3D-CNN, and HybridSN [3]. Transformer-based competitors include DCTN [17], HiT [16], and SSFTT [15]. The state-space group contains VMamba [19], MambaHSI [20], S$^2$Mamba [21], and WaveMamba [9]. Together, these methods cover local spectral–spatial extraction, attention-based dependency modeling, and state-space sequence representation [2], [45]. The same three evaluation metrics are used throughout the com-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



TABLE I: Classification performance (%) on the Houston2013 dataset using only 1% labeled samples for training. Results are reported as mean ± standard deviation over ten independent runs. The best result in each row is highlighted in bold.

| No. | Class Name | CNN-based | | | Transformer-based | | | Mamba-based | | | | SSCD-HI |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 2D-CNN | 3D-CNN | HybridSN | DCTN | HiT | SSFTT | VMamba | MambaHSI | $S^2$Mamba | WaveMamba | |
| 1 | Healthy Grass | 83.80 ± 10.60 | 72.34 ± 13.54 | 53.38 ± 13.87 | **90.52 ± 5.22** | 82.65 ± 6.52 | 85.17 ± 7.11 | 88.41 ± 7.34 | 86.99 ± 4.93 | 64.46 ± 15.28 | 75.74 ± 5.50 | 86.55 ± 4.42 |
| 2 | Stressed Grass | 88.58 ± 6.38 | 80.35 ± 9.77 | 83.52 ± 13.17 | 95.00 ± 2.85 | 92.19 ± 6.87 | 89.09 ± 9.84 | 95.45 ± 3.92 | 86.86 ± 8.01 | 95.32 ± 4.51 | **96.03 ± 2.72** | 91.18 ± 4.79 |
| 3 | Synthetic Grass | 86.85 ± 4.25 | 72.03 ± 12.30 | 94.70 ± 4.54 | 89.52 ± 4.23 | 53.29 ± 15.28 | 85.28 ± 5.51 | 88.33 ± 5.62 | 85.11 ± 6.28 | 66.62 ± 7.27 | 80.79 ± 4.96 | **98.43 ± 1.86** |
| 4 | Trees | 86.49 ± 5.83 | 81.14 ± 6.80 | 78.01 ± 5.57 | **99.77 ± 0.26** | 79.82 ± 3.25 | 81.90 ± 6.33 | 85.87 ± 3.19 | 82.45 ± 3.33 | 79.63 ± 2.88 | 83.08 ± 6.02 | 94.27 ± 2.93 |
| 5 | Soil | 97.40 ± 1.53 | 93.21 ± 6.36 | 84.89 ± 13.05 | 80.40 ± 6.31 | 97.24 ± 4.67 | 96.89 ± 2.70 | 98.93 ± 1.26 | 97.36 ± 3.51 | **99.15 ± 2.25** | 87.52 ± 10.14 | 97.80 ± 1.61 |
| 6 | Water | 47.39 ± 20.28 | 21.49 ± 19.44 | 22.64 ± 17.45 | 73.83 ± 6.66 | 27.83 ± 18.71 | 71.37 ± 12.13 | 78.76 ± 3.10 | 66.77 ± 4.79 | 47.95 ± 25.12 | 54.04 ± 19.58 | **82.12 ± 4.91** |
| 7 | Residential | 80.51 ± 9.99 | 63.02 ± 21.78 | 34.89 ± 30.52 | 77.68 ± 6.07 | 67.55 ± 7.28 | 92.25 ± 7.61 | 89.24 ± 4.08 | 88.24 ± 7.07 | **95.44 ± 3.72** | 86.53 ± 6.13 | 93.43 ± 2.56 |
| 8 | Commercial | 71.52 ± 7.14 | 59.66 ± 9.64 | 39.77 ± 17.33 | **83.95 ± 4.24** | 53.81 ± 6.87 | 63.41 ± 9.19 | 73.99 ± 7.25 | 70.56 ± 7.07 | 56.44 ± 15.81 | 50.99 ± 6.74 | 75.22 ± 8.21 |
| 9 | Road | 77.76 ± 6.28 | 68.26 ± 8.98 | 18.58 ± 18.47 | 86.86 ± 10.05 | 72.92 ± 6.90 | 82.86 ± 5.87 | 79.27 ± 6.28 | **89.96 ± 3.05** | 83.27 ± 4.40 | 83.47 ± 4.39 | 85.25 ± 5.24 |
| 10 | Highway | 86.85 ± 7.25 | 63.18 ± 14.52 | 62.73 ± 17.12 | 80.44 ± 8.84 | 67.22 ± 13.82 | 87.01 ± 6.67 | **95.79 ± 7.96** | 91.06 ± 9.35 | 72.33 ± 11.36 | 81.12 ± 10.45 | 94.06 ± 6.56 |
| 11 | Railway | 78.46 ± 11.01 | 38.32 ± 18.20 | 33.55 ± 23.66 | 81.26 ± 7.39 | 48.10 ± 26.35 | **90.91 ± 7.52** | 87.01 ± 6.98 | 86.87 ± 4.34 | 70.22 ± 13.02 | 88.23 ± 7.38 | 88.25 ± 6.10 |
| 12 | Parking Lot 1 | 77.25 ± 8.53 | 67.95 ± 11.90 | 44.05 ± 17.46 | 64.91 ± 4.93 | 71.76 ± 14.29 | 66.04 ± 9.38 | 82.65 ± 8.82 | 73.46 ± 6.84 | 54.43 ± 8.60 | 48.19 ± 10.84 | **87.09 ± 5.45** |
| 13 | Parking Lot 2 | 80.54 ± 10.39 | 21.51 ± 26.92 | 76.96 ± 26.96 | **96.77 ± 3.72** | 42.87 ± 18.45 | 81.66 ± 10.22 | 90.26 ± 4.77 | 72.28 ± 8.93 | 2.63 ± 7.40 | 75.13 ± 11.25 | 86.10 ± 5.34 |
| 14 | Tennis Court | 69.36 ± 23.21 | 27.29 ± 14.86 | 38.11 ± 16.64 | **97.37 ± 3.80** | 78.30 ± 11.73 | 85.57 ± 11.39 | 85.75 ± 11.58 | 88.18 ± 11.72 | 94.41 ± 3.93 | 72.15 ± 18.45 | 95.98 ± 3.35 |
| 15 | Running Track | 77.52 ± 10.18 | 47.40 ± 23.59 | 31.84 ± 17.40 | 85.56 ± 0.99 | 87.14 ± 6.98 | 93.46 ± 7.28 | 92.21 ± 11.58 | 91.91 ± 7.80 | 98.02 ± 2.55 | 75.99 ± 10.49 | **98.56 ± 2.51** |
| OA (%) | | 81.52 ± 1.91 | 64.00 ± 4.64 | 54.33 ± 5.79 | 85.56 ± 0.99 | 69.81 ± 2.79 | 83.68 ± 1.56 | 87.46 ± 1.11 | 84.86 ± 1.44 | 74.05 ± 2.72 | 76.81 ± 1.97 | **90.08 ± 0.70** |
| AA (%) | | 79.35 ± 3.17 | 58.48 ± 4.93 | 53.18 ± 6.64 | 85.59 ± 0.93 | 68.18 ± 2.88 | 83.53 ± 1.49 | 87.46 ± 1.24 | 83.87 ± 1.46 | 72.02 ± 3.22 | 75.93 ± 2.63 | **90.29 ± 0.77** |
| $\kappa \times 100$ | | 79.99 ± 2.08 | 60.90 ± 5.06 | 50.63 ± 6.33 | 84.38 ± 1.07 | 67.31 ± 3.02 | 82.36 ± 1.68 | 86.44 ± 1.21 | 83.62 ± 1.56 | 71.88 ± 2.95 | 74.93 ± 2.13 | **89.27 ± 0.75** |

TABLE II: Classification performance (%) on the Indian Pines dataset using 10% labeled samples for training. Results are reported as mean ± standard deviation over ten independent runs. The best result in each row is highlighted in bold.

| No. | Class Name | CNN-based | | | Transformer-based | | | Mamba-based | | | | SSCD-HI |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 2D-CNN | 3D-CNN | HybridSN | DCTN | HiT | SSFTT | MambaHSI | $S^2$Mamba | VMamba | WaveMamba | |
| 1 | Alfalfa | 35.12 ± 23.68 | 0.00 ± 0.00 | 1.71 ± 5.12 | 72.93 ± 13.06 | 20.98 ± 21.10 | 89.76 ± 9.56 | 67.07 ± 14.48 | 2.20 ± 3.53 | 87.32 ± 7.62 | 13.90 ± 16.40 | **97.07 ± 4.20** |
| 2 | Corn-notill | 91.67 ± 2.42 | 76.61 ± 6.61 | 72.12 ± 11.48 | 92.86 ± 2.12 | 89.87 ± 1.47 | 91.87 ± 1.65 | 89.45 ± 2.33 | 87.66 ± 2.86 | 92.51 ± 4.08 | 91.23 ± 3.26 | **96.78 ± 0.94** |
| 3 | Corn-mintill | 94.90 ± 1.46 | 68.15 ± 10.22 | 46.64 ± 31.35 | 94.38 ± 3.75 | 89.37 ± 10.30 | 94.47 ± 1.92 | 89.10 ± 5.26 | 88.51 ± 7.79 | 94.16 ± 4.29 | 89.05 ± 7.67 | **98.18 ± 1.09** |
| 4 | Corn | 87.23 ± 8.90 | 16.90 ± 12.76 | 36.76 ± 38.19 | 95.77 ± 2.80 | 89.30 ± 7.60 | 94.84 ± 8.76 | 95.63 ± 4.04 | 93.76 ± 5.69 | 92.39 ± 5.37 | 88.69 ± 8.89 | **99.91 ± 0.19** |
| 5 | Grass-pasture | 92.90 ± 2.38 | 72.87 ± 10.82 | 46.69 ± 38.96 | 93.95 ± 2.36 | 70.55 ± 7.37 | 94.14 ± 3.14 | 86.92 ± 6.78 | 82.94 ± 7.16 | 94.62 ± 3.68 | 91.43 ± 3.13 | **96.73 ± 2.26** |
| 6 | Grass-trees | 94.38 ± 1.17 | 86.09 ± 2.84 | 74.47 ± 29.20 | 96.93 ± 0.60 | 95.57 ± 1.05 | 94.34 ± 2.33 | 94.32 ± 1.07 | 93.73 ± 1.48 | 95.66 ± 1.79 | 93.64 ± 3.39 | **99.91 ± 0.16** |
| 7 | Grass-pasture-mowed | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 16.40 ± 28.42 | 0.00 ± 0.00 | 65.60 ± 35.15 | 0.00 ± 0.00 | 0.00 ± 0.00 | 67.20 ± 23.99 | 0.80 ± 2.40 | **100.00 ± 0.00** |
| 8 | Hay-windrowed | 99.77 ± 0.70 | 99.19 ± 0.93 | 69.70 ± 34.06 | 99.98 ± 0.07 | 99.51 ± 0.76 | **100.00 ± 0.00** | 99.74 ± 0.77 | **100.00 ± 0.00** | **100.00 ± 0.00** | 99.84 ± 0.23 | 99.98 ± 0.07 |
| 9 | Oats | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.00 ± 0.00 | 56.67 ± 17.71 | 0.00 ± 0.00 | 0.00 ± 0.00 | 61.67 ± 22.42 | 17.78 ± 20.31 | **100.00 ± 0.00** |
| 10 | Soybean-notill | 79.47 ± 2.97 | 62.11 ± 3.38 | 39.15 ± 36.65 | 80.72 ± 2.08 | 77.54 ± 2.66 | 82.55 ± 3.21 | 79.91 ± 2.51 | 80.99 ± 2.17 | 84.26 ± 3.45 | 79.82 ± 3.45 | **98.16 ± 1.45** |
| 11 | Soybean-mintill | 98.11 ± 1.43 | 90.17 ± 3.99 | 91.39 ± 3.72 | 98.25 ± 0.64 | 94.93 ± 1.74 | 98.14 ± 1.04 | 97.97 ± 0.49 | 97.49 ± 0.73 | 97.14 ± 2.00 | 97.03 ± 1.38 | **98.68 ± 0.42** |
| 12 | Soybean-clean | 93.63 ± 2.24 | 64.68 ± 8.03 | 45.96 ± 37.45 | 92.68 ± 3.61 | 90.39 ± 2.99 | 93.41 ± 2.45 | 93.33 ± 3.10 | 92.57 ± 4.89 | 95.13 ± 2.84 | 90.86 ± 5.78 | **96.34 ± 1.14** |
| 13 | Wheat | 83.68 ± 7.22 | 43.41 ± 21.71 | 19.46 ± 28.53 | 97.14 ± 2.54 | 88.59 ± 5.36 | 89.08 ± 4.83 | 95.14 ± 4.56 | 86.32 ± 8.48 | 93.62 ± 4.74 | 77.62 ± 13.22 | **99.95 ± 0.16** |
| 14 | Woods | 98.69 ± 1.01 | 98.10 ± 0.83 | 96.42 ± 4.72 | **99.91 ± 0.06** | 99.52 ± 0.44 | 99.02 ± 0.57 | 99.50 ± 0.49 | 99.53 ± 0.34 | 99.35 ± 0.41 | 98.95 ± 0.69 | 99.89 ± 0.16 |
| 15 | Buildings-Grass-Trees-Drives | 91.64 ± 4.12 | 46.05 ± 25.85 | 47.41 ± 35.04 | 92.13 ± 6.66 | 85.53 ± 15.40 | 88.65 ± 4.89 | 90.17 ± 4.99 | 89.16 ± 5.57 | 94.84 ± 3.49 | 88.56 ± 4.97 | **99.19 ± 0.59** |
| 16 | Stone-Steel-Towers | 30.83 ± 27.52 | 0.00 ± 0.00 | 0.00 ± 0.00 | 47.50 ± 10.16 | 40.12 ± 24.05 | 27.26 ± 30.69 | 29.05 ± 18.18 | 0.00 ± 0.00 | 35.12 ± 28.65 | 7.02 ± 16.76 | **99.88 ± 0.36** |
| OA (%) | | 92.43 ± 0.54 | 76.72 ± 2.59 | 67.78 ± 16.80 | 93.89 ± 0.46 | 89.66 ± 1.40 | 93.48 ± 0.46 | 91.95 ± 0.67 | 90.57 ± 0.93 | 94.07 ± 0.60 | 90.94 ± 1.03 | **98.48 ± 0.13** |
| AA (%) | | 73.25 ± 2.67 | 51.52 ± 4.17 | 42.99 ± 17.94 | 79.47 ± 1.57 | 70.74 ± 2.73 | 84.99 ± 2.71 | 75.46 ± 1.78 | 68.43 ± 1.08 | 86.56 ± 3.52 | 70.39 ± 2.24 | **98.79 ± 0.31** |
| $\kappa \times 100$ | | 91.34 ± 0.62 | 72.96 ± 3.12 | 62.03 ± 20.49 | 93.02 ± 0.52 | 88.17 ± 1.61 | 92.56 ± 0.52 | 90.79 ± 0.77 | 89.20 ± 1.08 | 93.23 ± 0.69 | 89.64 ± 1.19 | **98.26 ± 0.14** |

TABLE III: Classification performance (%) on the WHU-Hi-LongKou dataset using 0.1% labeled samples for training. Results are reported as mean ± standard deviation over ten independent runs. The highest mean in each row is highlighted in bold.

| No. | Class Name | CNN-based | | | Transformer-based | | | Mamba-based | | | | SSCD-HI |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 2D-CNN | 3D-CNN | HybridSN | DCTN | HiT | SSFTT | VMamba | MambaHSI | $S^2$Mamba | WaveMamba | |
| 1 | Corn | **99.58 ± 0.26** | 98.59 ± 0.67 | 99.00 ± 1.19 | 99.41 ± 1.13 | 99.13 ± 0.39 | 98.97 ± 0.34 | 99.06 ± 0.68 | 99.22 ± 0.64 | 98.24 ± 3.87 | 98.77 ± 0.82 | 98.73 ± 1.47 |
| 2 | Cotton | 92.50 ± 6.63 | 72.02 ± 13.73 | 82.54 ± 13.07 | 90.67 ± 13.43 | 81.04 ± 12.64 | 73.41 ± 26.74 | 92.26 ± 3.61 | 94.16 ± 4.18 | **95.63 ± 4.21** | 94.62 ± 2.86 | 82.58 ± 5.39 |
| 3 | Sesame | 76.58 ± 22.21 | 18.14 ± 25.71 | 44.62 ± 32.28 | 84.69 ± 19.98 | 2.89 ± 5.69 | 0.00 ± 0.00 | 80.83 ± 9.21 | 89.94 ± 5.04 | 7.06 ± 20.91 | 7.04 ± 18.66 | **91.33 ± 3.45** |
| 4 | Broad-leaf soybean | 99.13 ± 0.71 | 99.26 ± 0.92 | 99.12 ± 0.72 | 98.85 ± 0.79 | 98.60 ± 0.73 | **99.29 ± 0.80** | 98.98 ± 0.72 | 98.41 ± 1.21 | 98.04 ± 2.81 | 98.66 ± 2.70 | 98.33 ± 0.18 |
| 5 | Narrow-leaf soybean | 58.85 ± 17.62 | 39.18 ± 25.26 | 32.14 ± 27.97 | 72.14 ± 22.66 | 18.33 ± 13.98 | 35.39 ± 22.83 | 32.86 ± 17.37 | 61.69 ± 23.82 | 38.71 ± 33.60 | 65.56 ± 17.04 | **79.68 ± 8.56** |
| 6 | Rice | 93.72 ± 4.56 | 87.23 ± 4.91 | 96.26 ± 3.02 | **99.24 ± 0.35** | 95.54 ± 2.36 | 97.98 ± 1.70 | 94.97 ± 3.41 | 95.04 ± 3.43 | 96.02 ± 3.30 | 86.82 ± 9.37 | 94.02 ± 3.95 |
| 7 | Water | 99.93 ± 0.09 | 99.93 ± 0.10 | 99.98 ± 0.01 | 99.97 ± 0.04 | 99.99 ± 0.01 | **100.00 ± 0.01** | 99.98 ± 0.02 | 99.83 ± 0.29 | 99.64 ± 0.93 | 99.95 ± 0.12 | 99.63 ± 0.35 |
| 8 | Roads and houses | 55.54 ± 19.13 | 53.76 ± 14.01 | 56.79 ± 14.89 | 82.70 ± 11.82 | 76.01 ± 6.15 | 71.60 ± 14.14 | 79.21 ± 15.04 | 77.18 ± 6.74 | 64.10 ± 25.39 | 54.46 ± 18.38 | **84.52 ± 7.56** |
| 9 | Mixed weed | 26.59 ± 25.14 | 25.20 ± 20.80 | 28.39 ± 23.60 | 43.88 ± 18.62 | 43.23 ± 12.71 | 31.79 ± 15.77 | 55.60 ± 14.11 | 68.43 ± 17.28 | 44.75 ± 20.20 | 54.44 ± 13.89 | **81.20 ± 8.26** |
| OA (%) | | 94.36 ± 1.28 | 91.65 ± 1.15 | 93.09 ± 1.22 | 96.28 ± 1.48 | 93.00 ± 0.67 | 92.87 ± 1.27 | 95.40 ± 0.75 | 96.27 ± 0.70 | 93.29 ± 1.89 | 93.55 ± 1.18 | **96.53 ± 0.64** |
| AA (%) | | 78.05 ± 6.57 | 65.92 ± 5.80 | 70.99 ± 6.06 | 85.73 ± 5.59 | 68.31 ± 2.52 | 67.60 ± 4.08 | 81.53 ± 3.28 | 87.10 ± 3.14 | 71.35 ± 6.40 | 73.37 ± 2.87 | **90.00 ± 1.72** |
| $\kappa \times 100$ | | 92.47 ± 1.75 | 88.78 ± 1.61 | 90.77 ± 1.67 | 95.08 ± 1.98 | 90.67 ± 0.90 | 90.45 ± 1.73 | 93.91 ± 1.00 | 95.08 ± 0.93 | 91.06 ± 2.55 | 91.42 ± 1.55 | **95.43 ± 0.85** |

parison. The main tables summarize the adopted benchmark protocol. They do not by themselves establish equality of baseline augmentation, optimization steps, checkpoint selection, or tuning budgets. Architectural differences therefore prevent attributing the cross-method accuracy margins exclusively to the intervention mechanism.

### *B. Comparison With Representative HSI Classifiers*

Tables I, II, and III report the quantitative results. Rather than showing a uniform advantage on every individual category, SSCD-HI exhibits its clearest benefit at the aggregate level and on several classes for which competing models show substantial performance variation. This behavior is consistent with the intended role of the proposed representation regula-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



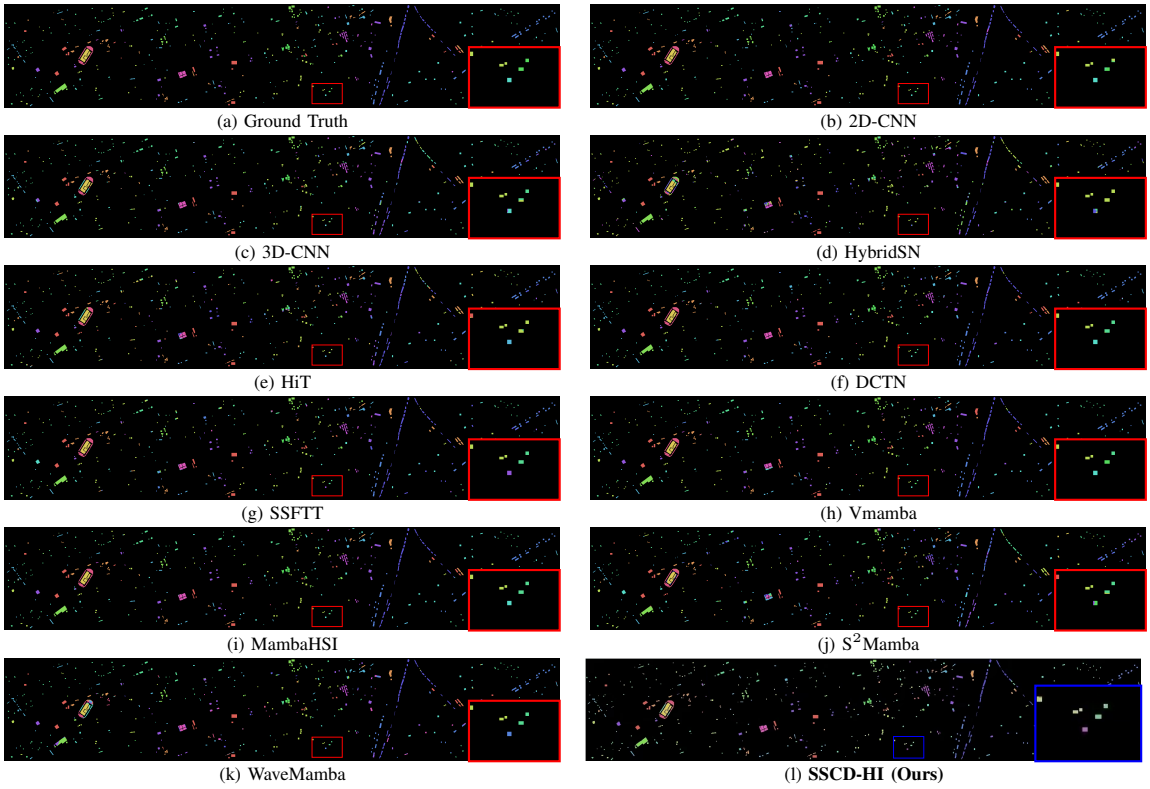



Fig. 3: Classification maps on Houston2013 (1% labeled samples).

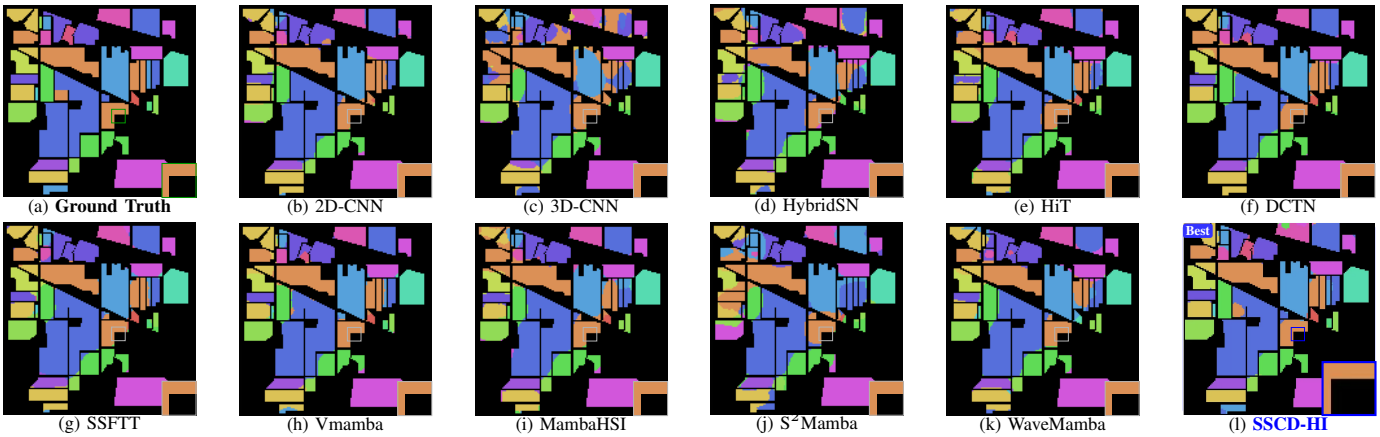



Fig. 4: Classification maps on the IP dataset (10% training samples).

tion: the method does not replace spectral–spatial feature extraction, but constrains how label-stable and nuisance-sensitive information contributes to the classifier.

*1) Results on Houston2013:* Houston2013 provides a challenging urban setting in which several materials share similar spectral responses and occur in heterogeneous neighborhoods. As shown in Table I, SSCD-HI achieves $90.08 \pm 0.70\%$ OA, $90.29 \pm 0.77\%$ AA, and $89.27 \pm 0.75$ in $\kappa \times 100$. The strongest competing OA in the table is obtained by VMamba at $87.46 \pm 1.11\%$. The corresponding differences in OA, AA, and $\kappa \times 100$ are 2.62, 2.83, and 2.83 percentage points, respectively.

The class-wise results give additional insight into this improvement. SSCD-HI reaches $82.12\%$ accuracy for Water, whereas the highest competing mean in that category is $78.76\%$. It also obtains $87.09\%$ for Parking Lot 1 and $98.43\%$ for Synthetic Grass. These categories are difficult for several convolutional and sequence-based baselines, as reflected by both lower means and relatively large standard deviations. The improvement is therefore not limited to a small change in the dominant urban classes.

The advantage is not uniform across all categories. DCTN remains stronger on Trees, Commercial, and Parking Lot 2, while other competitors obtain the highest accuracy on Road, Highway, or Railway. This observation is important: the aggregate advantage of SSCD-HI arises from a more balanced performance profile rather than from dominating every class independently. The relatively small standard deviations of its OA and AA also indicate stable aggregate behavior across the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



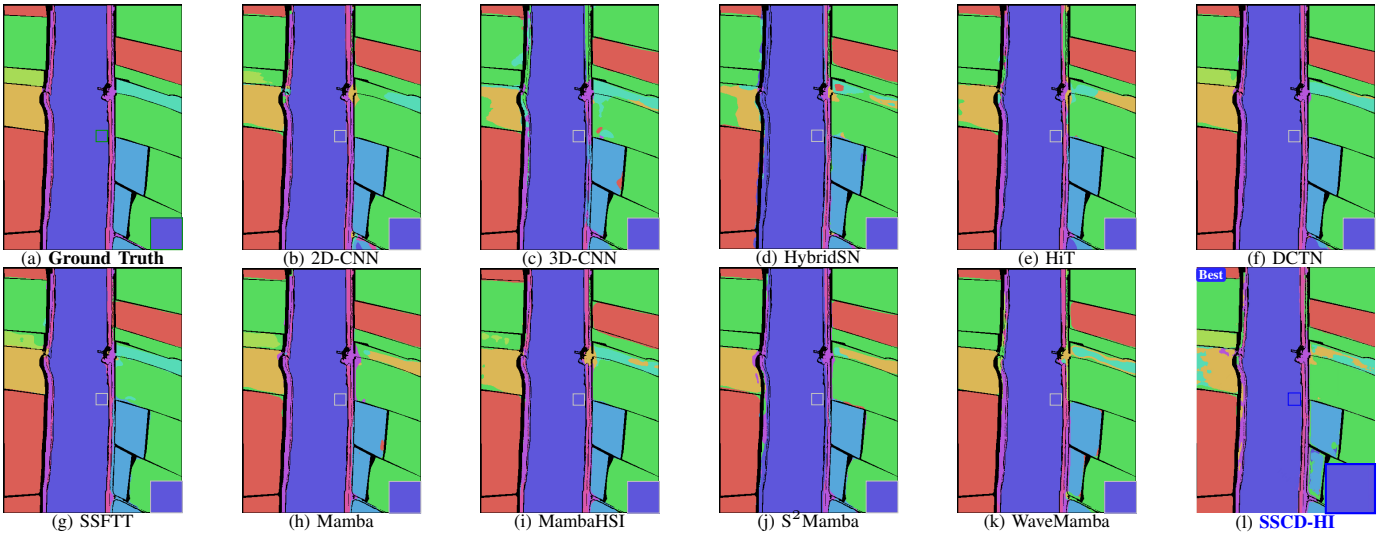



Fig. 5: Classification maps on the WHU-Hi-LongKou dataset (0.1% training samples).

reported runs.

*2) Results on Indian Pines:* Table II shows a similar pattern on Indian Pines. SSCD-HI obtains $98.48 \pm 0.13\%$ OA, $98.79 \pm 0.31\%$ AA, and $98.26 \pm 0.14$ in $\kappa \times 100$. The highest competing OA is $94.07 \pm 0.60\%$, leading to an OA difference of 4.41 percentage points. The difference is more pronounced for AA: the best baseline reaches $86.56\%$, compared with $98.79\%$ for SSCD-HI. The corresponding $\kappa \times 100$ difference is 5.03 points.

The considerably larger gain in AA than in OA is noteworthy. It indicates that the improvement is not concentrated only on categories that dominate the test set. Several classes that remain difficult for the compared methods are classified more accurately by SSCD-HI. For example, the reported mean accuracies for Alfalfa, Oats, and Stone-Steel-Towers are $97.07\%$, $100.00\%$, and $99.88\%$, respectively. In contrast, several competing methods show large variations or very low accuracy on these categories. The result is consistent with the purpose of regulating latent information rather than relying only on a high-capacity backbone.

For already well-separated categories, the margin is naturally smaller. Hay-windrowed and Woods, for example, are classified accurately by several baselines. This saturation explains why class-balanced measures provide additional information beyond OA on Indian Pines.

*3) Results on WHU-Hi-LongKou:* WHU-Hi-LongKou yields a different comparison because several methods already achieve high OA. SSCD-HI obtains $96.53 \pm 0.64\%$ OA, $90.00 \pm 1.72\%$ AA, and $95.43 \pm 0.85$ in $\kappa \times 100$. DCTN reaches the strongest competing OA of $96.28 \pm 1.48\%$, so the OA margin is only 0.25 percentage points, and MambaHSI follows closely at $96.27 \pm 0.70\%$. The difference in AA, however, is larger: $90.00\%$ for SSCD-HI versus $87.10\%$ for MambaHSI, the strongest baseline in that measure. The corresponding difference in $\kappa \times 100$ is 0.35 points.

The class-wise results explain this discrepancy between OA and AA. Categories such as Corn and Water are already close to saturation for several methods, leaving little room for additional OA improvement. More difficult categories show a clearer separation. SSCD-HI reaches $91.33\%$ on Sesame, $79.68\%$ on Narrow-leaf soybean, $84.52\%$ on Roads and houses, and $81.20\%$ on Mixed weed. These categories exhibit substantially lower accuracies for several baselines. The improvement in AA therefore reflects a more even performance across the scene rather than a large increase on classes that are already nearly perfectly classified.

SSCD-HI is again not the strongest method for every category. The highest Corn accuracy is obtained by 2D-CNN, whereas SSFTT attains the highest reported means on Broad-leaf soybean and Water. The result nevertheless shows that the proposed representation constraint remains competitive on saturated classes while providing larger gains on the more difficult ones.

### C. Qualitative Analysis

Figures 3, 4, and 5 present full-scene classification maps for the three benchmarks, complementing the quantitative results with spatial error distributions. On Houston2013, SSCD-HI produces cleaner homogeneous regions, fewer isolated misclassifications, and more distinct class boundaries than most competing methods, consistent with its improvements in OA and AA reported in Table I. Similar behavior is observed on Indian Pines and WHU-Hi-LongKou, particularly in regions containing spectrally similar or spatially heterogeneous classes. Overall, the qualitative results support the quantitative analysis and indicate that regulating label-stable and nuisance-sensitive latent information contributes to more spatially coherent HSI classification.

### D. Ablation Study

Table IV distinguishes the main-protocol component results from additional validation-selected controls. The Complete rows in both panels reproduce the main-table OA, AA, and kappa values. Panel B includes this row solely as a main-protocol reference, not as a validation-selected observation. default patch, label-ratio, and zero-noise results reuse the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



TABLE IV: Component ablations and additional controls (mean ± SD, OA and AA in percent). Complete rows reproduce the main results. Other rows in Panel B use separate validation-selected retraining (ten paired seeds, sample SD), their matched full reference is reported in the response letter.

**A. Manuscript protocol**

| | Components | | | Indian Pines | | | Houston2013 | | | WHU-Hi-LongKou | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Configuration | Mask | Align. | HI | OA | AA | $100\kappa$ | OA | AA | $100\kappa$ | OA | AA | $100\kappa$ |
| Without masks | ✗ | ✓ | ✓ | 97.86±0.16 | 96.36±0.43 | 97.56±0.18 | 85.54±3.16 | 84.63±4.33 | 84.35±3.44 | 95.04±0.89 | 83.39±3.04 | 93.45±1.18 |
| Without alignment | ✓ | ✗ | ✓ | 97.85±0.13 | 93.47±1.97 | 97.55±0.15 | 84.49±2.29 | 83.08±3.27 | 83.21±2.48 | 95.07±0.73 | 84.45±1.80 | 93.50±0.95 |
| Without HI | ✓ | ✓ | ✗ | 97.80±0.39 | 94.89±1.77 | 97.49±0.45 | 85.22±1.41 | 84.37±2.17 | 84.01±1.53 | 94.83±1.14 | 82.72±3.45 | 93.17±1.52 |
| Complete | ✓ | ✓ | ✓ | **98.48±0.13** | **98.79±0.31** | **98.26±0.14** | **90.08±0.70** | **90.29±0.77** | **89.27±0.75** | **96.53±0.64** | **90.00±1.72** | **95.43±0.85** |

**B. Additional controls**

| | Indian Pines | | | Houston2013 | | | WHU-Hi-LongKou | | |
|---|---|---|---|---|---|---|---|---|---|
| Configuration | OA | AA | $100\kappa$ | OA | AA | $100\kappa$ | OA | AA | $100\kappa$ |
| Complete (main reference) | 98.48±0.13 | 98.79±0.31 | 98.26±0.14 | 90.08±0.70 | 90.29±0.77 | 89.27±0.75 | 96.53±0.64 | 90.00±1.72 | 95.43±0.85 |
| Without masks | 96.72±0.62 | 94.15±2.71 | 96.26±0.71 | 85.44±1.27 | 85.68±1.72 | 84.26±1.38 | 94.05±1.69 | 82.77±3.63 | 92.18±2.20 |
| Complementary masks | 96.58±0.58 | 93.18±1.64 | 96.10±0.66 | 83.63±2.44 | 83.74±2.90 | 82.29±2.65 | 94.15±1.38 | 82.56±5.46 | 92.29±1.82 |
| Without HI | 96.71±0.76 | 92.66±2.16 | 96.25±0.87 | 84.40±2.91 | 84.41±3.27 | 83.13±3.15 | 94.61±0.89 | 83.52±2.91 | 92.89±1.17 |
| Without alignment | 96.54±0.68 | 93.09±2.76 | 96.06±0.78 | 84.82±1.53 | 84.98±1.87 | 83.58±1.65 | 94.81±0.60 | 84.02±2.58 | 93.16±0.79 |
| Without HI/alignment | 96.41±0.59 | 92.30±2.12 | 95.91±0.67 | 84.05±2.68 | 83.89±3.66 | 82.75±2.91 | 92.80±2.10 | 81.57±3.25 | 90.58±2.67 |
| Intra-class only | 95.99±1.16 | 92.61±3.30 | 95.43±1.33 | 85.14±1.96 | 85.24±2.44 | 83.94±2.13 | 94.13±1.01 | 81.94±4.10 | 92.25±1.34 |
| Cross-class only | 96.52±0.51 | 93.13±2.18 | 96.03±0.58 | 83.60±3.13 | 83.94±3.19 | 82.27±3.39 | 93.43±0.82 | 79.91±4.72 | 91.35±1.08 |
| Trained stable branch | 96.36±0.65 | 92.13±1.69 | 95.85±0.74 | 83.42±2.58 | 83.82±2.21 | 82.07±2.79 | 93.82±1.30 | 82.04±2.30 | 91.87±1.68 |
| Trained nuisance branch | 96.82±0.40 | 92.65±1.75 | 96.37±0.46 | 82.22±3.50 | 82.50±3.37 | 80.78±3.79 | 93.90±1.34 | 81.61±2.69 | 91.97±1.72 |

TABLE V: Patch size ablation study on three HSI benchmarks. Values are mean ± standard deviation over ten random runs.

| Patch Size | Indian Pines | | | Houston2013 | | | WH | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | $\kappa$ | OA | AA | $\kappa$ | OA | AA | $\kappa$ |
| 7 | $96.85 \pm 0.25$ | $91.73 \pm 0.47$ | $96.47 \pm 0.36$ | $83.49 \pm 1.36$ | $83.09 \pm 2.77$ | $81.49 \pm 1.59$ | $95.26 \pm 0.69$ | $83.26 \pm 3.46$ | $93.80 \pm 0.90$ |
| 11 | $97.78 \pm 0.12$ | $93.77 \pm 1.21$ | $97.53 \pm 0.18$ | $84.26 \pm 1.25$ | $83.42 \pm 3.02$ | $82.29 \pm 1.46$ | $95.08 \pm 1.02$ | $83.29 \pm 3.04$ | $93.58 \pm 1.33$ |
| 15 | $98.48 \pm 0.13$ | $98.79 \pm 0.31$ | $98.26 \pm 0.14$ | $90.08 \pm 0.70$ | $90.29 \pm 0.77$ | $89.27 \pm 0.75$ | $96.53 \pm 0.64$ | $90.00 \pm 1.72$ | $95.43 \pm 0.85$ |
| 19 | $97.53 \pm 0.25$ | $93.77 \pm 1.21$ | $97.27 \pm 0.37$ | $84.26 \pm 2.03$ | $82.72 \pm 3.90$ | $82.27 \pm 2.37$ | $95.40 \pm 0.73$ | $85.29 \pm 3.67$ | $93.99 \pm 0.94$ |
| 23 | $97.37 \pm 0.34$ | $92.48 \pm 1.15$ | $97.01 \pm 0.45$ | $82.67 \pm 2.43$ | $81.72 \pm 4.37$ | $80.47 \pm 2.84$ | $95.02 \pm 0.37$ | $83.67 \pm 1.68$ | $93.53 \pm 0.48$ |

main configuration. The validation-selected controls reserve 20% of the original training-label budget for validation and selects checkpoints by validation (OA+AA)/2. The controls use patch size 15, batch size 100, AdamW with learning rate 0.001 and weight decay 0.0001, and 100/200/200 epochs for Indian Pines/Houston2013/WHU-Hi-LongKou. The validation-selected full reference is reported in the response letter and supplies the matched baseline for these controls and the default coefficient pair.

The validation-selected results do not reproduce universal component gains. Removing masks increases OA by 0.31 and 1.74 percentage points on Indian Pines and Houston2013, respectively, but decreases it by 0.24 points on WHU-Hi-LongKou. Removing HI increases OA by 0.30, 0.70, and 0.31 points, respectively. Removing alignment increases it by 0.14, 1.12, and 0.51 points. These descriptive differences qualify the original component-gain interpretation: the benefit depends on the training and model-selection protocol. Complementary masks and separately trained single branches are included as controls, rather than inferred from zeroing a frozen joint-head input. Neither protocol establishes a class-free nuisance branch or physically valid latent recombination.

### *E. Sensitivity to Spatial Context*

Table V reports the sensitivity of SSCD-HI to the input patch size. Among the evaluated settings, $15 \times 15$ consistently provides the best OA, AA, and kappa on all three datasets. Specifically, the corresponding OA values are $98.48\%$, $90.08\%$, and $96.53\%$ on Indian Pines, Houston2013, and WHU-Hi-LongKou, respectively.

Both smaller and larger spatial windows lead to lower performance. On Houston2013, for example, OA increases from $83.49\%$ at $7 \times 7$ to $90.08\%$ at $15 \times 15$, but decreases to $82.67\%$ when the patch size is enlarged to $23 \times 23$. Similar behavior is observed on Indian Pines and WHU-Hi-LongKou. These results suggest that the $15 \times 15$ neighborhood provides an effective balance between local spectral evidence and spatial context within the tested range. It is therefore adopted as the default patch size in the remaining experiments.

### *F. Sensitivity to Training Ratio*

Table VI evaluates the sensitivity of SSCD-HI to the amount of labeled training data. Performance generally improves as additional supervision becomes available, although the magnitude of the improvement is dataset dependent.

On Houston2013, the default ratio of 0.01 yields $90.08\%$ OA and $90.29\%$ AA, while increasing the ratio to 0.10 raises

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



TABLE VI: Sensitivity of SSCD-HI to different training ratios on three HSI benchmarks. Results are reported as mean ± standard deviation over repeated runs.

| Dataset | Training Ratio ($r$) | OA (%) | AA (%) | $\kappa \times 100$ |
|---|---|---|---|---|
| Houston2013 | 0.005 | 83.57±4.09 | 83.64±3.86 | 82.23±4.42 |
| | 0.01 | 90.08±0.70 | 90.29±0.77 | 89.27±0.75 |
| | 0.02 | 92.97±0.42 | 93.15±0.65 | 92.40±0.46 |
| | 0.05 | 97.09±0.51 | 96.89±0.58 | 96.86±0.55 |
| | 0.10 | 98.72±0.27 | 98.80±0.32 | 98.61±0.30 |
| Indian Pines | 0.01 | 75.44±4.35 | 63.04±3.86 | 72.02±4.82 |
| | 0.05 | 94.75±0.59 | 87.83±1.79 | 94.02±0.68 |
| | 0.10 | 98.48±0.13 | 98.79±0.31 | 98.26±0.14 |
| | 0.15 | 98.64±0.29 | 98.22±0.68 | 98.45±0.33 |
| | 0.20 | 99.04±0.25 | 99.00±0.29 | 98.91±0.29 |
| WHU-Hi-LongKou | 0.001 | 96.53±0.64 | 90.00±1.72 | 95.43±0.85 |
| | 0.002 | 97.42±0.16 | 91.81±0.88 | 96.60±0.22 |
| | 0.005 | 98.68±0.05 | 96.34±0.22 | 98.26±0.07 |
| | 0.01 | 99.37±0.04 | 98.42±0.25 | 99.17±0.05 |
| | 0.02 | 99.59±0.02 | 99.09±0.08 | 99.46±0.03 |

TABLE VII: Noise perturbation study on three HSI benchmarks. Values are mean ± standard deviation over ten random runs. OA and AA are percentages, kappa is reported as $\kappa \times 100$.

| Noise std. | Indian Pines | | | Houston2013 | | | WH | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | $\kappa \times 100$ | OA | AA | $\kappa \times 100$ | OA | AA | $\kappa \times 100$ |
| 0.00 | 98.48 ± 0.13 | 98.79 ± 0.31 | 98.26 ± 0.14 | 90.08 ± 0.70 | 90.29 ± 0.77 | 89.27 ± 0.75 | 96.53 ± 0.64 | 90.00 ± 1.72 | 95.43 ± 0.85 |
| 0.05 | 94.23 ± 0.29 | 90.40 ± 1.32 | 93.67 ± 0.40 | 78.10 ± 1.34 | 76.49 ± 2.53 | 75.47 ± 1.56 | 94.85 ± 0.57 | 83.79 ± 1.91 | 93.20 ± 0.74 |
| 0.10 | 79.28 ± 2.58 | 73.76 ± 4.60 | 76.15 ± 3.13 | 52.52 ± 4.99 | 55.33 ± 3.58 | 51.81 ± 5.74 | 94.77 ± 0.53 | 82.90 ± 2.98 | 93.06 ± 0.69 |
| 0.15 | 60.64 ± 5.99 | 50.91 ± 6.65 | 54.72 ± 7.11 | 34.23 ± 3.79 | 37.36 ± 2.04 | 30.98 ± 4.35 | 94.30 ± 0.69 | 82.72 ± 4.21 | 92.52 ± 0.90 |
| 0.20 | 47.83 ± 6.16 | 36.45 ± 5.42 | 38.62 ± 6.70 | 26.06 ± 5.79 | 30.86 ± 5.41 | 21.56 ± 6.53 | 92.94 ± 0.38 | 79.79 ± 3.72 | 90.96 ± 0.50 |

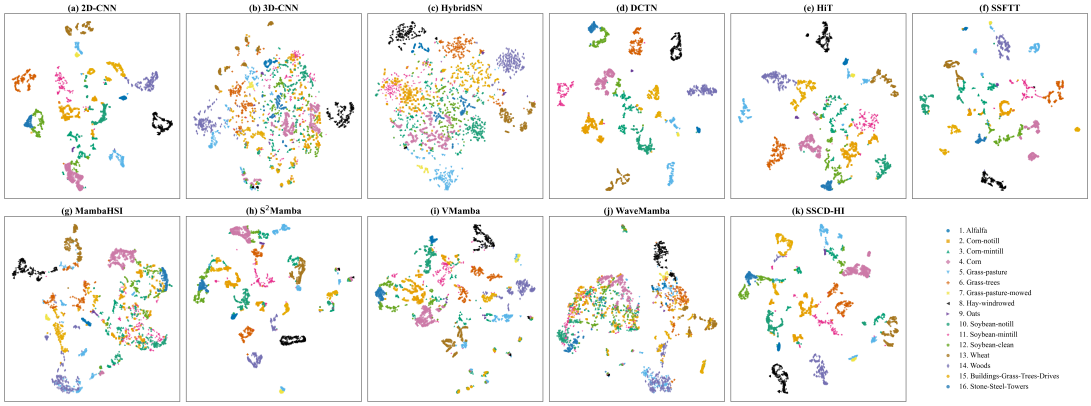



Fig. 6: t-SNE visualization of learned representations on Indian Pines: (a) 2D-CNN, (b) 3D-CNN, (c) HybridSN, (d) DCTN, (e) HiT, (f) SSFTT, (g) MambaHSI, (h) S$^2$Mamba, (i) VMamba, (j) WaveMamba, and (k) SSCD-HI. All panels contain the same 2764 stratified test samples and share identical class encodings and t-SNE settings. Panels (h) and (i) are one-epoch reference-backend diagnostics rather than converged baseline results.

these values to 98.72% and 98.80%, respectively. Indian Pines exhibits a stronger dependence on supervision density, with OA increasing from 75.44% at 0.01 to 99.04% at 0.20. In contrast, WHU-Hi-LongKou already reaches 96.53% OA at the default ratio of 0.001 and improves steadily as more labeled samples are added, reaching 99.59% OA at 0.02.

These results indicate that SSCD-HI generally benefits from additional labeled samples, but the magnitude of the gain depends on scene characteristics. Indian Pines and Houston2013 profit most from increased supervision, whereas WHU-Hi-LongKou already attains a high accuracy at the sparsest setting and improves more gradually thereafter.

### G. Noise Perturbation Analysis

Table VII examines the sensitivity of SSCD-HI to increasing spectral perturbation. Such degradation can affect the reliability of HSI classification [46]. The results show a clear dataset-dependent response rather than uniform tolerance to noise.

Indian Pines and Houston2013 are increasingly affected as the perturbation level rises. For Indian Pines, OA decreases from 98.48% without added noise to 79.28% at a standard deviation of 0.10 and 47.83% at 0.20. Houston2013 exhibits a stronger decline, with OA falling from 90.08% to 52.52% and 26.06% at the same two noise levels. In contrast, WHU-Hi-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



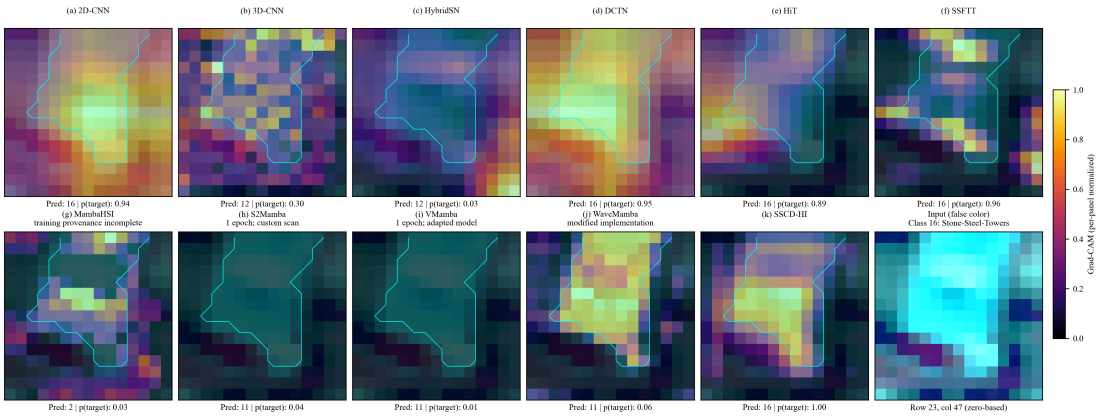



Fig. 7: Grad-CAM visualization for a selected Stone-Steel-Towers patch from Indian Pines, centered at row 23 and column 47 (zero-based). All maps explain the ground-truth class logit and are normalized independently to $[0, 1]$. Cyan contours delineate pixels annotated as the target class. The displayed probabilities correspond to that class.

LongKou remains comparatively stable, retaining $92.94\%$ OA and a kappa value above 90 at the highest perturbation level.

These differences indicate that the sensitivity of SSCD-HI depends strongly on the spectral–spatial characteristics of the scene. The pronounced degradation on Indian Pines and Houston2013 under severe perturbation also defines a practical limitation of the current model. Accordingly, this experiment should be interpreted as a within-model sensitivity analysis rather than evidence of superior noise robustness. Establishing such an advantage would require competing methods to be evaluated under the same perturbation protocol.

### *H. Visualization Analysis*

Fig. 6 compares the penultimate representations of ten baselines with the label-stable representation $\mathbf{z}^c$ learned by SSCD-HI on Indian Pines. To ensure a controlled comparison, all panels use the same 2764 stratified test samples, class colors, marker shapes, random seed, and t-SNE hyperparameters. The SSCD-HI embedding forms multiple compact class-conditional manifolds with limited overlap for several spectrally similar crop categories. DCTN, HiT, SSFTT, and 2D-CNN also exhibit well-separated clusters, whereas 3D-CNN, MambaHSI, and WaveMamba retain more mixed regions. Because t-SNE is a nonlinear projection whose geometry is sensitive to optimization and hyperparameters, this visualization is treated as qualitative evidence and is interpreted together with the quantitative classification results. The $S^2$Mamba and VMamba panels are implementation diagnostics obtained with a one-epoch portable reference-scan run. They are included to verify feature extraction but are excluded from claims about converged representation quality.

### *I. Class Activation Analysis*

Fig. 7 presents a selected Stone-Steel-Towers patch from Indian Pines. In this example, SSCD-HI concentrates its positive Grad-CAM response within the annotated target region, whereas 2D-CNN and DCTN exhibit broader spatial responses. The target region contains 68.3% of the total positive activation for SSCD-HI, compared with 38.6%, 40.0%, 43.7%, and 57.4% for 2D-CNN, DCTN, HiT, and SSFTT, respectively. SSCD-HI also predicts the correct class. 3D-CNN and HybridSN assign the patch to Soybean-clean. For this example, the activation distribution indicates closer spatial alignment with the target annotation. Independently normalized maps and a single selected patch do not provide an aggregate comparison across models.

TABLE VIII: Indian Pines coefficient settings (mean $\pm$ SD). OA and AA are percentages.

| $\lambda_{\mathrm{HI}}$ | $\lambda_{\mathrm{dec}}$ | OA | AA | $100\kappa$ |
|---|---|---|---|---|
| 0.1 | 0.1 | 96.44±0.67 | 92.97±1.59 | 95.94±0.76 |
| 0.5 | 0.01 | 96.18±0.85 | 92.89±1.75 | 95.64±0.98 |
| 0.5 | 0.1 | 98.48±0.13 | 98.79±0.31 | 98.26±0.14 |
| 0.5 | 0.2 | 96.90±0.37 | 93.04±1.97 | 96.47±0.42 |
| 1 | 0.1 | 96.55±0.84 | 93.71±1.87 | 96.07±0.96 |

### *J. Validation-Selected Loss-Weight Sensitivity*

Table VIII reports the main-protocol reference and four one-factor-at-a-time validation-selected controls on Indian Pines. The default $(0.5, 0.1)$ row reproduces the Indian Pines main result in Table II. The other four rows use independent validation-selected retraining and are not matched ablations of that main result. The four controls vary one coefficient at a time around the validation-selected default. They do not identify an optimal main-protocol pair.

## V. Conclusion

We proposed SSCD-HI to regulate the predictive use of spectral–spatial latent information under limited labels. Adaptive masks produce two projected components, while intra-class nuisance exchange, cross-class nuisance intervention, and an angular alignment penalty shape their use during training. The intervention paths reuse encoded features and are removed at inference. The reported experiments on Indian Pines, Houston2013, and WHU-Hi-LongKou assess classification performance, component contributions, spatial context, label

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



availability, and spectral perturbation. Branch readouts show residual class information in both components, while separately trained controls qualify the component-gain interpretation. This distinction places the contribution in intervention-guided prediction regularization rather than identified physical causal disentanglement.

## REFERENCES


[1] Y. Song, J. Zhang, Z. Liu, Y. Xu, S. Quan, L. Sun, J. Bi, and X. Wang, "Deep learning for hyperspectral image classification: A comprehensive review and future predictions," *Information Fusion*, vol. 123, p. 103285, 2025.

[2] M. Ahmad, S. Distefano, A. M. Khan, M. Mazzara, C. Li, H. Li, J. Aryal, Y. Ding, G. Vivone, and D. Hong, "A comprehensive survey for hyperspectral image classification: The evolution from conventional to transformers and mamba models," *Neurocomputing*, vol. 644, p. 130428, 2025.

[3] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "Hybridsn: Exploring 3-d–2-d cnn feature hierarchy for hyperspectral image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 17, no. 2, pp. 277–281, 2020.

[4] D. Hong, Z. Han, J. Yao, L. Gao, B. Zhang, A. Plaza, and J. Chanussot, "Spectralformer: Rethinking hyperspectral image classification with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022.

[5] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022.

[6] S. Mohamed, M. Haghighat, T. Fernando, S. Sridharan, C. Fookes, and P. Moghadam, "Factoformer: Factorized hyperspectral transformers with self-supervised pretraining," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–14, 2024.

[7] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3dss-mamba: 3d-spectral–spatial mamba for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[8] X. Tang, Y. Yao, J. Ma, X. Zhang, Y. Yang, B. Wang, and L. Jiao, "Spiralmamba: Spatial-spectral complementary mamba with spatial spiral scan for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–19, 2025.

[9] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "Wavemamba: Spatial-spectral wavelet mamba for hyperspectral image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, p. 5500505, 2025.

[10] H. Wang, X. Liu, Z. Qiao, and H. Tao, "Inducing causal meta-knowledge from virtual domain: Causal meta-generalization for hyperspectral domain generalization," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[11] D. Peng, J. Wu, T. Han, Y. Li, Y. Wen, G. Yang, and L. Qu, "Disentanglement-inspired single-source domain-generalization network for cross-scene hyperspectral image classification," *Knowledge-Based Systems*, vol. 303, p. 112413, 2024.

[12] X. Li, Z. Yang, and W. Li, "Causehsi: Counterfactual-augmented domain generalization for hyperspectral image classification via causal disentanglement," *Journal of Imaging*, vol. 12, no. 2, p. 57, 2026.

[13] Y. Liu, Z. Shen, H. Yang, W. Liu, and X. Yang, "Cat: Causal attention with linear complexity for efficient and interpretable hyperspectral image classification," *Remote Sensing*, vol. 18, no. 2, p. 358, 2026.

[14] F. Zhao, J. Zhang, Z. Meng, and H. Liu, "Densely connected pyramidal dilated convolutional network for hyperspectral image classification," *Remote Sensing*, vol. 13, no. 17, p. 3396, 2021.

[15] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.

[16] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.

[17] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "Dctn: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[18] Y. Xu, D. Wang, L. Zhang, and L. Zhang, "Dual selective fusion transformer network for hyperspectral image classification," *Neural Networks*, vol. 187, p. 107311, 2025.

[19] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, J. Jiao, and Y. Liu, "Vmamba: Visual state space model," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[20] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "Mambahsi: Spatial–spectral mamba for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[21] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S$^2$mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–13, 2025.

[22] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, C. Richard, J. Chanussot, L. Drumetz, J.-Y. Tourneret, A. Zare, and C. Jutten, "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geoscience and Remote Sensing Magazine*, vol. 9, no. 4, pp. 223–270, 2021.

[23] J. M. Bioucas-Dias, A. Plaza, N. Dobigeon, M. Parente, Q. Du, P. Gader, and J. Chanussot, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 5, no. 2, pp. 354–379, 2012.

[24] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[25] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," *arXiv:1907.02893*, 2019.

[26] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009.

[27] B. Schölkopf, F. Locatello, S. Bauer, N. R. Ke, N. Kalchbrenner, A. Goyal, and Y. Bengio, "Toward causal representation learning," *Proc. IEEE*, vol. 109, no. 5, pp. 612–634, 2021.

[28] Y. Huang, J. Peng, W. Sun, N. Chen, Q. Du, Y. Ning, and H. Su, "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2022.

[29] Y. Huang, J. Peng, W. Sun, N. Chen, Z. Ye, and Q. Du, "Domain-aware adversarial domain augmentation network for hyperspectral image classification," *IEEE Transactions on Image Processing*, vol. 35, pp. 1275–1289, 2026.

[30] X. Wang, S. Dong, X. Zheng, R. Lu, and J. Jia, "Explicit high-level semantic network for domain generalization in hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–14, 2024.

[31] C. Lin, Z. You, and Y. Lin, "CINet: Causal inference inspired network for cross-scene hyperspectral image classification," *Journal of Physics: Conference Series*, vol. 3083, p. 012006, 2025.

[32] W. Zha, L. Yang, H. Xu, B. Wang, D. Chen, Y. Lin, Y. Wang, and P. S.-P. Wang, "Multi-scale adaptive diffusion feature fusion with causal invariance learning for hyperspectral image classification," *International Journal of Pattern Recognition and Artificial Intelligence*, vol. 39, no. 06, p. 2550010, 2025.

[33] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016.

[34] H. Zhao, H. Yu, L. Lin, Z. Zhang, S. Gao, and J. Wang, "Asymmetric disentanglement for domain generalization in hyperspectral image classification," *Neurocomputing*, p. 132461, 2025.

[35] H. Zhao, L. Lin, J. Wang, S. Gao, and Z. Zhang, "Adversarial decoupling domain generalization network for cross-scene hyperspectral image classification," *Knowledge-Based Systems*, vol. 318, p. 113432, 2025.

[36] T. Wang, C. Zhou, Q. Sun, and H. Zhang, "Causal attention for unbiased visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 3091–3100.

[37] F. Locatello, S. Bauer, M. Lucic, G. Rätsch, S. Gelly, B. Schölkopf, and O. Bachem, "Challenging common assumptions in the unsupervised learning of disentangled representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 4114–4124.

[38] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[39] V. Verma, A. Lamb, C. Beckham, A. Najafi, I. Mitliagkas, A. Courville, D. Lopez-Paz, and Y. Bengio, "Manifold mixup: Better representations by interpolating hidden states," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 97, 2019, pp. 6438–6447.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3748457



[40] A. Gretton, O. Bousquet, A. Smola, and B. Schölkopf, "Measuring statistical dependence with Hilbert–Schmidt norms," in *Proc. Int. Conf. Algorithmic Learn. Theory (ALT)*, 2005, pp. 63–77.
[41] G. J. Székely, M. L. Rizzo, and N. K. Bakirov, "Measuring and testing dependence by correlation of distances," *The Annals of Statistics*, vol. 35, no. 6, pp. 2769–2794, 2007.
[42] D. Li, F. Kong, and D. Zhu, "Hyperspectral image classification based on local gaussian mixture feature extraction," *Acta Optica Sinica*, vol. 41, no. 6, p. 0610001, 2021.
[43] Z. Xu, W. Jiang, and J. Geng, "Texture-aware causal feature extraction network for multimodal remote sensing data classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–12, 2024.
[44] X. Guo, Q. Feng, and F. Guo, "Cmtnet: a hybrid cnn-transformer network for uav-based hyperspectral crop classification in precision agriculture," *Scientific Reports*, vol. 15, no. 1, p. 12383, 2025.
[45] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Computer Science Review*, vol. 53, p. 100658, 2024.
[46] C. Li, Z. Li, X. Liu, and S. Li, "The influence of image degradation on hyperspectral image classification," *Remote Sensing*, vol. 14, no. 20, p. 5199, 2022.



**Xiaofei Yang** Xiaofei Yang received the B.S. degree from Suihua University, Suihua, China in 2011, and received M.S. and Ph.D. degrees from Harbin Institute of Technology, China, in 2014 and 2019, respectively. He was a Postdoctoral with the Department of Computer and Information Science, University of Macau, China, from 2020 to 2023. He is currently a lecturer with School of Electronics and Communications Engineering, Guangzhou University.

His research interests are in the areas of semi-supervised learning, deep learning, remote sensing, transfer learning and graph mining.



**Tong Tang** Tong Tang is currently pursuing the B.Eng. degree in Communication Engineering with the School of Electronic and Communication Engineering, Guangzhou University, Guangzhou, China. Her research interests include hyperspectral image classification and state-space models for spectral–spatial feature propagation.



**Ying Liu** Ying Liu received the B.Sc. (Hons.), M.Ph., and Ph.D. degrees in telecommunications engineering with examiners-commendation from the University of Wollongong, Wollongong, NSW, Australia, in 2012, 2015, and 2021, respectively. She was a Postdoctoral Fellow with the School of Software Engineering, Sun YatSen University, Zhuhai, China. She is currently a Lecturer with the School of Electronic and Communication Engineering,Guangzhou University, Guangzhou, China.

Her research interests include resource allocation in IoTs, backscatter communications, Deep Learning, and blockchain in Internet of Things.



**Yuanchao Su** (Senior Member, IEEE) received the Ph.D. degree from Sun Yat-sen University, Guangzhou, China, in 2019. His current research interests include deep learning, data fusion, hyperspectral unmixing, remote sensing image processing, and graph neural networks.

He was awarded the 2023 Macau Young Scholars Program (MYSP). Moreover, he was also awarded the Shaanxi Province Youth Science and Technology Star. From 2024 to 2026, he joined the Department of Computer and Information Science, University of Macau, as a MYSP Postdoctoral Fellow. From 2018 to 2019, he was a Visiting Researcher with the Department of Electrical Engineering and Computer Science, University of Tennessee at Knoxville, Knoxville, TN, USA. Since 2019, he has been serving at the Department of Remote Sensing, College of Geomatics, Xi'an University of Science and Technology, where he is an Associate Professor and Doctoral Supervisor.

Dr. Su has been a Senior Member of the IEEE Geoscience and Remote Sensing Society since 2022. He also serves as a Reviewer and Guest Editor for some international journals.



**Yifang Ban** Yifang Ban (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from Nanjing University, Nanjing, China, in 1984 and 1987, respectively, and the Ph.D. degree from the University of Waterloo, Waterloo, ON, Canada, in 1996. Before joining KTH Royal Institute of Technology (KTH), Stockholm, Sweden in 2004, she was a tenured Associate Professor at York University, Toronto, ON, Canada. She is currently the Chair Professor and the Director of the Geoinformatics Division at and an Associate Director of Digital Futures in Stockholm. Her research interests include Earth observation big data analytics, machine learning/deep learning and their applications in mapping urban and land cover, monitoring urbanization, wildfires, and other environmental changes, and assessing environmental impact. She has published extensively on these topics.

Dr. Ban is the Co-Chair of the ICA Commission on Sensor-Driven Mapping and a Co-Lead of the Group on Earth Observations (GEO) initiative "Global Urban Observation and Information" (2012-2022). She has been an Associate Editor and a Guest Editor for major remote sensing journals and an Invited Expert for EU and national grant application evaluations. Since 2016, she has been an Invited Expert of the UN Habitat Technical Committee on Human Settlements Indicators for UN Sustainable Development Goals (SDGs).



**Yicong Zhou** Yicong Zhou (M'07-SM'14) received his Ph.D. degree in electrical engineering from Tufts University, Medford, MA, USA. He is a Professor in the Department of Computer and Information Science at University of Macau, Macau, China. His research interests include image processing, computer vision, machine learning, and multimedia security.

Dr. Zhou is a Fellow of SPIE (the Society of Photo-Optical Instrumentation Engineers) and was recognized as one of "Highly Cited Researchers" in 2020, 2021, 2023 and 2024. He serves as a Senior Area Editor for IEEE Transactions on Circuits and Systems for Video Technology and an Associate Editor for IEEE Transactions on Cybernetics, IEEE Transactions on Neural Networks and Learning Systems, and IEEE Transactions on Geoscience and Remote Sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:18:47 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.